
O-RAN Work Group 11 (Security Work Group)

OAuth 2.0 Security

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Foreword

This Technical Report (TR) has been produced by WG11 of the O-RAN ALLIANCE.

The content of the present document is subject to continuing work within O-RAN and may change following formal O-RAN approval. Should the O-RAN Alliance modify the contents of the present document, it will be re-released by O-RAN with an identifying change of version date and an increase in version number as follows:

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Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in O-RAN deliverables except when used in direct citation.

1 Scope

The technical report defines the authorization framework for O-RAN architecture elements using OAuth 2.0 and applies to the HTTP and Kafka based APIs [i.13] [i.18] used in the R1, O1, O2, A1, FH m-plane [i.17] and Y1 interfaces.

For this objective, the present document

- Defines the procedure flow for token registration process, token request process, token verification process and token authorization process. It will specify the token scope metadata for O-RAN.
- Analyses the existing specifications related to OAuth 2.0 such as O-RAN Security protocol specification, IETF RFC 8705 [i.6], IETF RFC 6749 [i.5], 3GPP TS29.501 [i.3], 3GPP TS 33.122 [i.4], 3GPP TS 28.319 [i.19] and ETSI NFV SEC-022 [i.7]. This is used as input to the present document.
- Analyses the security threat arising from the misuse of the access token and defines the security requirements associated to access token.
- Provides the study and recommendation of an authorization server architecture to overlay on the O-RAN architecture.

2 References

2.1 Normative references

Not applicable.

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18 or the latest 3GPP release prior to Release 18 that includes that document.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long term validity.

The following referenced documents are not necessary for the application of the present document but they assist the user with regard to a particular subject area.

- [i.1] O-RAN ALLIANCE TS: "O-RAN Security Protocols Specification"
- [i.2] O-RAN ALLIANCE TS: "O-RAN Security Requirements and Controls Specification"
- [i.3] 3GPP TS 29.501: "5G System; Network Function Repository Services"
- [i.4] 3GPP TS 33.122: "Security aspects of Common API Framework (CAPIF) for 3GPP northbound APIs"
- [i.5] IETF RFC 6749: "The OAuth 2.0 Authorization Framework"
- [i.6] IETF RFC 8705: "OAuth 2.0 Mutual-TLS Client Authentication and Certificate-Bound Access Tokens"
- [i.7] ETSI GS NFV-SEC 022: "Network Functions Virtualisation (NFV) Release 2; Security; Access Token Specification for API Access"
- [i.8] IETF RFC 6819: "OAuth 2.0 Threat Model and Security Considerations"
- [i.9] O-RAN ALLIANCE TS: "O-RAN Architecture Description"
- [i.10] 3GPP TS 23.222: "Common API Framework for 3GPP Northbound APIs"
- [i.11] 3GPP TS 29.222: "Common API Framework for 3GPP Northbound APIs"
- [i.12] IETF RFC 2616: "Hypertext Transfer Protocol -- HTTP/1.1"

- [i.13] O-RAN TS: "R1 interface: General Aspects and Principles" ("R1GAP")
- [i.14] O-RAN TS: "Non-RT RIC & A1/R1 Interface: Use Cases and Requirements" ("UCR")
- [i.15] O-RAN TR: "Decoupled SMO"
- [i.16] O-RAN TR: "SMO Security"
- [i.17] O-RAN TS: "Management Plane Specification"
- [i.18] Kafka Documentation <https://kafka.apache.org/documentation.html>
- [i.19] 3GPP TS 29.319: "Management and orchestration; Access control for management services"
- [i.20] O-RAN ALLIANCE TR: "O-RAN Threat Modelling and Risk Assessment"
- [i.21] Final: OpenID Connect Core 1.0 incorporating errata set 2
- [i.22] IETF RFC 7591: OAuth 2.0 Dynamic Client Registration Protocol
- [i.23] IETF RFC 8414: OAuth 2.0 Authorization Server Metadata
- [i.24] ITU-T M.3050: <https://www.kornev-online.net/ITIL/T-REC-M.3050-200405-I!Sup3!PDF-E.pdf>
- [i.25] [Coarse-Grained vs Fine-Grained Authorization: Which to Use?](#)

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the following terms apply:

A1: Interface between non-RT RIC and Near-RT RIC to enable policy-driven guidance of Near-RT RIC applications/functions, and support AI/ML workflow.

Access Token: credentials used to obtain access. It is a string, usually opaque to the client, representing an authorization issued to the client. Tokens represent specific scopes and durations of access, granted by the resource owner, and enforced by the resource server and authorization server (see [i.5], clause 1.4).

Authorization code: The authorization code is obtained by using an authorization server as an intermediary between the client and resource owner. Instead of requesting authorization directly from the resource owner, the client directs the resource owner to an authorization server (via its user-agent as defined in [IETF RFC 2616]), which in turn directs the resource owner back to the client with the authorization code. ([i.5], clause 1.3.1)

Authorization Server: The server issuing access tokens to the client after successfully authenticating the resource owner and obtaining authorization. ([i.5], clause 1.1)

Client: An application making protected resource requests on behalf of the resource owner and with its authorization. The term "client" does not imply any particular implementation characteristics (e.g., whether the application executes on a server, a desktop, or other devices). ([i.5], clause 1.1)

Client Credentials: The client credentials (or other forms of client authentication) can be used as an authorization grant when the authorization scope is limited to the protected resources under the control of the client, or to protected resources previously arranged with the authorization server. Client credentials are used as an authorization grant typically when the client is acting on its own behalf (the client is also the resource owner) or is requesting access to protected resources based on an authorization previously arranged with the authorization server. ([i.5], clause 1.3.4)

Coarse-grained authorization (CGA): uses one attribute (like user roles), making access control simple for smaller setups (see [i.25]).

Confidential client: A client capable of maintaining the confidentiality of their credentials (e.g., client implemented on a secure server with restricted access to the client credentials), or capable of secure client authentication using other means as in clause 2.2 of [i.5].

CSRF: Cross-site request forgery (CSRF) is a web-based attack whereby HTTP requests are transmitted from a user that the web site trusts or has authenticated (e.g., via HTTP redirects or HTML forms). ([i.8], clause 4.4.1.8)

Fine-grained authorization (FGA): uses a combination of detailed attributes, offering precise control for larger, intricate environments (see [i.25]).

Near-Real-Time RAN Intelligent Controller: An O-RAN Network Function (NF) that enables near-real-time control and optimization of RAN elements and resources via fine-grained data collection and actions over E2 interface. It may include AI/ML (Artificial Intelligence / Machine Learning) workflow including model training, inference and updates.

Non-Real-Time RAN Intelligent Controller: A functionality within SMO that drives the content carried across the A1 interface. It is comprised of the Non-RT RIC Framework and the Non-RT RIC Applications (rApps) whose services are defined below.

Non-RT RIC Applications (rApps): Modular applications that leverage the functionality exposed via the Non-RT RIC Framework's R1 interface to provide added value services relative to RAN operation, such as driving the A1 interface, recommending values and actions that may be subsequently applied over the O1/O2 interface and generating "enrichment information" for the use of other rApps. The rApp functionality within the Non-RT RIC enables non-real-time control and optimization of RAN elements and resources and policy-based guidance to the applications/features in Near-RT RIC.

O2: Interface between SMO framework as specified in clause **Error! Reference source not found.** and the O-Cloud for supporting O-RAN virtual network functions. Please refer to **Error! Reference source not found.** for more information.

Open Redirector: An open redirector is an endpoint using a parameter to automatically redirect a user-agent to the location specified by the parameter value without any validation. ([i.5], clause 10.15)

R1 Interface: Interface between rApps and Non-RT RIC Framework via which R1 Services can be produced and consumed.

Refresh token: Refresh tokens are credentials used to obtain access tokens. Refresh tokens are issued to the client by the authorization server and are used to obtain a new access token when the current access token becomes invalid or expires, or to obtain additional access tokens with identical or narrower scope (access tokens may have a shorter lifetime and fewer permissions than authorized by the resource owner). ([i.5], clause 1.5)

Resource Owner: An entity capable of granting access to a protected resource. When the resource owner is a person, it is referred to as an end-user. ([i.5], clause 1.1)

Resource Server: The server hosting the protected resources, capable of accepting and responding to protected resource requests using access tokens. ([i.5], clause 1.1)

SMO: A Service Management and Orchestration framework.

User-Agent: The User-Agent request-header field contains information about the user agent originating the request. This is for statistical purposes, the tracing of protocol violations, and automated recognition of user agents for the sake of tailoring responses to avoid particular user agent limitations. ([i.12], clause 14.43)

xApp: An application designed to run on the Near-RT RIC. Such an application is likely to consist of one or more microservices and at the point of on-boarding will identify which data it consumes and which data it provides. The application is independent of the Near-RT RIC and may be provided by any third party. The E2 enables a direct association between the xApp and the RAN functionality.

Y1: An interface over which RAN analytics services are exposed by the Near-RT RIC to be consumed by Y1 consumers.

3.2 Symbols

Void

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

3GPP	3rd Generation Partnership Project
Near-RT RIC	Near-Real-Time RAN Intelligent Controller
rApp	Non-RT RIC Application
SMO	Service Management and Orchestration
xApp	Near-RT RIC Application

4 Study and recommendation for OAuth 2.0

4.1 General Consideration

This Technical Report makes the following considerations:

- O-RAN Alliance WG1 has identified the key functions and interfaces [i.9] adopted in O-RAN and this will influence the set of assets to be protected for OAuth 2.0 framework.
- This TR will perform a detailed study of the OAuth 2.0 authorization server for O-RAN architecture elements and interfaces.
- Clause 4.7 in O-RAN Security Protocol Security specification [i.1] will be considered as the baseline text for the OAuth 2.0 framework.
- The recommended security requirements and controls provided in this report will be shared with the impacted O-RAN Alliance working group, such as WG1, WG2, WG3, WG6 and WG10 prior publishing them as normative.

4.2 IETF 6749

Editor's note: This clause will do the study related to OAuth 2.0 framework from IETF RFC 6749

4.3 IETF 8705

Editor's note: This clause will do the study related to OAuth 2.0 Mutual-TLS Client Authentication and Certificate-Bound Access Tokens from IETF RFC 8705

4.4 3GPP TS 33.122

Editor's note: This clause will do the study related to OAuth 2.0 recommended from 3GPP specification (23.222, 29.222, 33.122, TS29.501)

4.5 ETSI NFV SEC022

Editor's note: This clause will do the study related to OAuth 2.0 mechanism and definition from ETSI NFV SEC022

5 Threats

5.1 Assets

The following O-RAN architecture elements and interfaces should be considered for the security threat analysis:

5.1.1 O-RAN Architecture Elements

- ASSET-C-01: Service Management and Orchestration (SMO)
- ASSET-C-02: Near-RT RIC software
- ASSET-C-07: O-eNB
- ASSET-C-08: O-Cloud
- ASSET-C-09: xApps
- ASSET-C-10: rApps
- ASSET-C-11: Non-RT RIC
- ASSET-C-32: O-RU
- ASSET-C-34: O-DU
- ASSET-C-36: O-CU Host, includes O-CU-CP and O-CU-UP software

5.1.2 O-RAN interfaces

- ASSET-C-14: A1 interface
- ASSET-C-16: R1 interface
- ASSET-C-23: O2 interface
- ASSET-C-42: Y1 interface
- ASSET-C-63: Near RT RIC API's

5.2 Threats Template

Template to present the threat characteristics:

Threat ID	O-RAN threat ID
Threat title	[Threat Title will include suitable reference of 3GPP/IETF/ETSI]
Threat description	Introduction about the threat
Threat type	Spoofing Tampering Repudiation Information disclosure Denial of Service Elevation of Privilege
Vulnerability	
Impact type	Authenticity Integrity Non-repudiation Confidentiality Availability Authorization
Affected Assets	Asset identification

5.3 Potential Threat and Exploits

5.4 OAuth 2.0 Threats

The following threats to OAuth 2.0 have been identified and are analyzed in clause 6 Threat Analysis.

Table 5.4-1 – OAuth 2.0 threats

Threat-Id	Threat Description (Brief)
General OAuth 2.0 Threats	
T-OAuth-G-01	Obtaining client secrets
T-OAuth-G-02	Obtaining refresh tokens
T-OAuth-G-03	Obtaining access tokens
T-OAuth-G-04	Eavesdropping access tokens
T-OAuth-G-05	Disclosure of client credentials during transmission
T-OAuth-G-06	Open redirectors on client
T-OAuth-G-07	Obtain a deployment specific secret
T-OAuth-G-08	Obtaining authorization by code substitution (OAuth Login)
T-OAuth-G-09	Access token leak in transport/endpoints
T-OAuth-G-10	Malicious client obtains existing authorization by fraud
T-OAuth-G-11	Malicious client obtains authorization
T-OAuth-G-12	Accidental exposure of passwords at client site
T-OAuth-G-13	Client obtains scopes without end-user authorization and possible misconfigurations
T-OAuth-G-14	Client obtains refresh token through automatic authorization
T-OAuth-G-15	Eavesdropping refresh tokens from authorization server
T-OAuth-G-16	Obtaining refresh token by online guessing
T-OAuth-G-17	Refresh token phishing by counterfeit authorization server
T-OAuth-G-18	Guessing access tokens
T-OAuth-G-19	Access token phishing by counterfeit resource server
T-OAuth-G-20	Abuse of token by legitimate resource server or client
T-OAuth-G-21	Token leakage via log files and http referrers
T-OAuth-G-22	User unintentionally grants too much access scope
T-OAuth-G-23	Leak of confidential data in HTTP proxies
T-OAuth-G-24	Replay of authorized resource server requests
Authorization Server related threats	
T-OAuth-AS-01	Password phishing by counterfeit authorization server
T-OAuth-AS-02	Obtaining access tokens from authorization server database
T-OAuth-AS-03	Obtaining client secret from authorization server database
T-OAuth-AS-04	Online guessing of authorization codes
T-OAuth-AS-05	DoS attacks that exhaust resources
T-OAuth-AS-06	CSRF attack against redirect-uri
T-OAuth-AS-07	Resource owner impersonation

T-OAuth-AS-08	DoS using manufactured authorization "codes"
T-OAuth-AS-09	Obtaining user passwords from authorization server database
T-OAuth-AS-10	Obtaining the resource owner password credentials by online guessing
T-OAuth-AS-11	Obtaining refresh token from authorization server database
Access Control related Threats	
T-O-RAN-03	Attacks from the internet exploit weak authentication and access control to penetrate O-RAN network boundary
T-O-RAN-06	An attacker exploits insufficient/improper mechanisms for authentication and authorization to compromise O-RAN components
T-R1-01	An attacker gains unauthorized access to R1 services
T-R1-05	An attacker gains unauthorized access to data
T-AppLCM-04	Attacker exploits missing or improperly defined elements of application's SecurityDescriptor
T-GEN-02	Malicious access to exposed services using valid accounts
T-SMO-10	Sensitive data at rest is exposed to an internal attacker

NOTE: In above table 5.4-1, access control related threats are defined in [i.20]

6 Threat Analysis

6.1 General OAuth 2.0 Threats

Threat ID	T-OAuth-G-01
Threat title	Obtaining client secrets, IETF RFC 6819 [i.8], clause 4.1.1
Threat description	A malicious actor may be access the secrets and token of O-RAN architecture elements and use them to replay its refresh tokens and authorization codes.
Threat type	Information disclosure Elevation of privilege
Vulnerability	Unsecured tokens and secrets
Impact type	Confidentiality Authorization
Affected Asset	ASSET-C-01, ASSET-C-02, ASSET-C-07, ASSET-C-08, ASSET-C-09, ASSET-C-10, ASSET-C-11, ASSET-C-32, ASSET-C-34, ASSET-C-36

Threat ID	T-OAuth-G-02
Threat title	Obtaining refresh tokens, IETF RFC 6819 [i.8], clause 4.1.2
Threat description	A malicious actor may get file system access of the local storage of O-RAN architecture elements and read the refresh tokens. A malicious actor may use the compromised refresh token for a malicious application.
Threat type	Information disclosure
Vulnerability	A new refresh token was not issued
Impact type	Confidentiality

Affected Asset	ASSET-C-01, ASSET-C-02, ASSET-C-07, ASSET-C-08, ASSET-C-09, ASSET-C-10, ASSET-C-11, ASSET-C-32, ASSET-C-34, ASSET-C-36
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Threat ID	T-OAuth-G-03
Threat title	Obtaining access tokens, IETF RFC 6819 [i.8], clause 4.1.3
Threat description	Access tokens may be stolen by a malicious actor from the local storage if the O-RAN architecture elements store them in a local storage that is accessible to other O-RAN architecture elements or applications without restriction proper access control.
Threat type	Elevation of privilege
Vulnerability	Insufficient protection for credentials
Impact type	Authorization
Affected Asset	ASSET-C-01, ASSET-C-02, ASSET-C-07, ASSET-C-08, ASSET-C-09, ASSET-C-10, ASSET-C-11, ASSET-C-32, ASSET-C-34, ASSET-C-36

Threat ID	T-OAuth-G-04
Threat title	Eavesdropping access tokens, IETF RFC 6819 [i.8], clause 4.3.1
Threat description	A malicious actor may attempt to eavesdrop on access tokens in transit from the authorization server to the O-RAN architecture elements and applications
Threat type	Information disclosure
Vulnerability	Non authenticated, weakly or incorrectly authenticated communication channels
Impact type	Confidentiality
Affected Asset	ASSET-C-01, ASSET-C-02, ASSET-C-07, ASSET-C-08, ASSET-C-09, ASSET-C-10, ASSET-C-11, ASSET-C-32, ASSET-C-34, ASSET-C-36

Threat ID	T-OAuth-G-05
Threat title	Disclosure of client credentials during transmission, IETF RFC 6819 [i.8], clause 4.3.3
Threat description	A malicious actor may attempt to eavesdrop the transmission of O-RAN architecture elements credentials between the application and authorization server during the O-RAN architecture elements authentication process or during OAuth token requests.
Threat type	Information disclosure
Vulnerability	Non authenticated, weakly or incorrectly authenticated communication channels
Impact type	Confidentiality
Affected Asset	ASSET-C-01, ASSET-C-02, ASSET-C-07, ASSET-C-08, ASSET-C-09, ASSET-C-10, ASSET-C-11, ASSET-C-32, ASSET-C-34, ASSET-C-36

Threat ID	T-OAuth-G-06
Threat title	Open redirectors on client, IETF RFC 6819 [i.8], clause 4.1.5

Threat description	A malicious actor may use an open redirector operated by the O-RAN Applications to construct a redirect URI that will pass the authorization server validation but will send authorization code to an endpoint under the control of a malicious actor.
Threat type	Information disclosure
Vulnerability	Non authenticated, weakly or incorrectly authenticated communication channels
Impact type	Authenticity
Affected Asset	ASSET-C-01, ASSET-C-02, ASSET-C-07, ASSET-C-08, ASSET-C-09, ASSET-C-10, ASSET-C-11, ASSET-C-32, ASSET-C-34, ASSET-C-36

Threat ID	T-OAuth-G-07
Threat title	Obtain a deployment-specific secret, IETF RFC 6819 [i.8], clause 4.1.1
Threat description	An attacker may try to gain the secret for a O-RAN client installation, either from a resource server or O-RAN Applications.
Threat type	Spoofing
Vulnerability	Malicious applications, insufficient access control
Impact type	Authenticity
Affected Asset	ASSET-C-01, ASSET-C-02, ASSET-C-07, ASSET-C-08, ASSET-C-09, ASSET-C-10, ASSET-C-11, ASSET-C-32, ASSET-C-34, ASSET-C-36

Threat ID	T-OAuth-08
Threat title	Obtaining authorization by code substitution (OAuth Login), IETF RFC 6819 [i.8], clause 4.4.1.13
Threat description	An attacker could attempt to log into an application using a victim's identity. Applications relying on identity data provided by an OAuth protected service API to login users are vulnerable to this threat.
Threat type	Information disclosure
Vulnerability	Insufficient protection for credentials
Impact type	Authorization
Affected Asset	ASSET-C-01, ASSET-C-02, ASSET-C-07, ASSET-C-08, ASSET-C-09, ASSET-C-10, ASSET-C-11, ASSET-C-32, ASSET-C-34, ASSET-C-36

Threat ID	T-OAuth-G-09
Threat title	Access token leak in transport/endpoints, IETF RFC 6819 [i.8], clause 4.4.2.1
Threat description	A token may be eavesdropped by an attacker. The token is sent from the server to the client via a URI fragment of the redirect URI. If the communication is not secured or the endpoint is not secured, the token could be leaked by parsing the returned URI.
Threat type	Information disclosure
Vulnerability	Non authenticated, weakly or incorrectly authenticated URI
Impact type	Authorization

Affected Asset	ASSET-C-01, ASSET-C-02, ASSET-C-07, ASSET-C-08, ASSET-C-09, ASSET-C-10, ASSET-C-11, ASSET-C-32, ASSET-C-34, ASSET-C-36
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Threat ID	T-OAuth-G-10
Threat title	Malicious client obtains existing authorization by fraud, IETF RFC 6819 [i.8], clause 4.2.3
Threat description	Authorization servers may wish to automatically process authorization requests from client that have been previously authorized by the user. When the user is redirected to the authorization server's end-user authorization endpoint to grant access, the authorization server detects that the user has already granted access to that particular client. Instead of prompting the user for approval, the authorization server automatically redirects the user back to the client. A malicious client may exploit that feature and try to obtain such an authorization "code" instead of the legitimate client.
Threat type	Spoofing Repudiation
Vulnerability	Improper authorization process and lack of re-validating the authorization requests
Impact type	Authorization Non repudiation
Affected Asset	ASSET-C-01, ASSET-C-02, ASSET-C-07, ASSET-C-08, ASSET-C-09, ASSET-C-10, ASSET-C-11, ASSET-C-32, ASSET-C-34, ASSET-C-36

Threat ID	T-OAuth-G-11
Threat title	Malicious client obtains authorization, IETF RFC 6819 [i.8], clause 4.4.2.3
Threat description	A malicious client may attempt to obtain a token by fraud.
Threat type	Spoofing Repudiation
Vulnerability	Insufficient protection for credentials
Impact type	Authorization Non repudiation
Affected Asset	ASSET-C-01, ASSET-C-02, ASSET-C-07, ASSET-C-08, ASSET-C-09, ASSET-C-10, ASSET-C-11, ASSET-C-32, ASSET-C-34, ASSET-C-36

Threat ID	T-OAuth-G-12
Threat title	Accidental exposure of passwords at client site, IETF RFC 6819 [i.8], clause 4.4.3.1
Threat description	If the client does not provide enough protection, an attacker could retrieve the passwords for a user.
Threat type	Information Disclosure
Vulnerability	Insufficient protection for credentials
Impact type	Authorization
Affected Asset	ASSET-C-01, ASSET-C-02, ASSET-C-07, ASSET-C-08, ASSET-C-09, ASSET-C-10, ASSET-C-11, ASSET-C-32, ASSET-C-34, ASSET-C-36

Threat ID	T-OAuth-G-13
Threat title	Client obtains scopes without end-user authorization and possible misconfigurations, IETF RFC 6819 [i.8], clause 4.4.3.2
Threat description	An O-RAN Application like an rApp or xApp intentionally or unintentionally obtains a token with scope unknown for, or unintended by, the resource owner.
Threat type	Tampering Elevation of privilege
Vulnerability	misconfiguration
Impact type	Confidentiality Authorization
Affected Asset	ASSET-C-01, ASSET-C-02, ASSET-C-07, ASSET-C-08, ASSET-C-09, ASSET-C-10, ASSET-C-11, ASSET-C-32, ASSET-C-34, ASSET-C-36

Threat ID	T-OAuth-G-14
Threat title	Client obtains refresh token through automatic authorization, IETF RFC 6819 4.4.3.3 [i.8], clause 4.4.3.3
Threat description	An O-RAN application like an rApp or xApp obtains intentionally or unintentionally a long-term authorization represented by a refresh token even if the resource owner did not intend so
Threat type	Tampering Elevation of privilege
Vulnerability	Improper authorization process and lack of re-validating the authorization requests
Impact type	Confidentiality Authorization
Affected Asset	ASSET-C-01, ASSET-C-02, ASSET-C-07, ASSET-C-08, ASSET-C-09, ASSET-C-10, ASSET-C-11, ASSET-C-32, ASSET-C-34, ASSET-C-36

Threat ID	T-OAuth-G-15
Threat title	Eavesdropping refresh tokens from authorization server, IETF RFC 6819 [i.8], clause 4.5.1
Threat description	An O-RAN Application like an rApp or xApp may eavesdrop on refresh tokens when they are transmitted from the authorization server to the client.
Threat type	Tampering Elevation of privilege
Vulnerability	Non authenticated, weakly or incorrectly authenticated communication channels
Impact type	Confidentiality Authorization
Affected Asset	ASSET-C-01, ASSET-C-02, ASSET-C-07, ASSET-C-08, ASSET-C-09, ASSET-C-10, ASSET-C-11, ASSET-C-32, ASSET-C-34, ASSET-C-36

Threat ID	T-OAuth-G-16
Threat title	Obtaining refresh token by online guessing, IETF RFC 6819 [i.8], clause 4.5.3
Threat description	An O-RAN Application like an rApp or xApp may try to guess valid refresh token values and send it using the grant type "refresh_token" in order to obtain a valid access token.

Threat type	Tampering Elevation of privilege
Vulnerability	Improper authorization process and lack of re-validating the authorization requests
Impact type	Confidentiality Authorization
Affected Asset	ASSET-C-01, ASSET-C-02, ASSET-C-07, ASSET-C-08, ASSET-C-09, ASSET-C-10, ASSET-C-11, ASSET-C-32, ASSET-C-34, ASSET-C-36

Threat ID	T-OAuth-G-17
Threat title	Refresh token phishing by counterfeit authorization server, IETF RFC 6819 [i.8], clause 4.5.4
Threat description	An O-RAN Application like an rApp or xApp could obtain valid refresh tokens by proxying requests to the authorization server.
Threat type	Tampering Spoofing Elevation of privilege
Vulnerability	Non authenticated, weakly or incorrectly authenticated communication channels
Impact type	Confidentiality Authorization
Affected Asset	ASSET-C-01, ASSET-C-02, ASSET-C-07, ASSET-C-08, ASSET-C-09, ASSET-C-10, ASSET-C-11, ASSET-C-32, ASSET-C-34, ASSET-C-36

Threat ID	T-OAuth-G-18
Threat title	Guessing access tokens, IETF RFC 6819 [i.8], clause 4.6.3
Threat description	Access to a single user's data is possible where an attacker may place a malicious rApp or xApp and attempt to guess the access token values based on knowledge from other access tokens.
Threat type	Tampering Elevation of privilege
Vulnerability	Insecure generation of access token
Impact type	Confidentiality Authorization
Affected Asset	ASSET-C-01, ASSET-C-02, ASSET-C-07, ASSET-C-08, ASSET-C-09, ASSET-C-10, ASSET-C-11, ASSET-C-32, ASSET-C-34, ASSET-C-36

Threat ID	T-OAuth-G-19
Threat title	Access token phishing by counterfeit resource server, IETF RFC 6819 [i.8], clause 4.6.4
Threat description	An O-RAN Application like an rApp or xApp may send a valid access token to a counterfeit resource server where the server in turn uses that token to access other services on behalf of the resource owner.
Threat type	Tampering Spoofing Elevation of privilege
Vulnerability	Insufficient protection for credentials
Impact type	Confidentiality Authorization

Affected Asset	ASSET-C-01, ASSET-C-02, ASSET-C-07, ASSET-C-08, ASSET-C-09, ASSET-C-10, ASSET-C-11, ASSET-C-32, ASSET-C-34, ASSET-C-36
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Threat ID	T-OAuth-G-20
Threat title	Abuse of token by legitimate resource server or client, IETF RFC 6819 [i.8], clause 4.6.5
Threat description	An O-RAN legitimate resource server which is protecting the resources could use an access token to access another resource server.
Threat type	Tampering Elevation of privilege
Vulnerability	insufficient protection for access control on resource server
Impact type	Confidentiality Authorization
Affected Asset	ASSET-C-01, ASSET-C-02, ASSET-C-07, ASSET-C-08, ASSET-C-09, ASSET-C-10, ASSET-C-11, ASSET-C-32, ASSET-C-34, ASSET-C-36

Threat ID	T-OAuth-G-21
Threat title	Token leakage via log files and HTTP referrers, IETF RFC 6819 [i.8], clause 4.6.7
Threat description	An O-RAN Application such as an rApp or xApp may incorrectly send the access token in the URI query, allowing the access tokens to be leaked in log files and the HTTP “referrer.”
Threat type	Tampering Elevation of privilege
Vulnerability	Improper implementation of API request
Impact type	Confidentiality Authorization
Affected Asset	ASSET-C-01, ASSET-C-02, ASSET-C-07, ASSET-C-08, ASSET-C-09, ASSET-C-10, ASSET-C-11, ASSET-C-32, ASSET-C-34, ASSET-C-36

Threat ID	T-OAuth-G-22
Threat title	User unintentionally grants too much access scope, IETF RFC 6819 [i.8], clause 4.2.2
Threat description	An O-RAN Application such as an rApp or xApp may incorrectly implement scoping and end-user authorization, allowing unauthorized access to resources.
Threat type	Elevation of privilege
Vulnerability	Misconfiguration, insufficient authorization access
Impact type	Authorization
Affected Asset	ASSET-C-01, ASSET-C-02, ASSET-C-07, ASSET-C-08, ASSET-C-09, ASSET-C-10, ASSET-C-11, ASSET-C-32, ASSET-C-34, ASSET-C-36

Threat ID	T-OAuth-G-23
Threat title	Leak of confidential data in HTTP proxies, IETF RFC 6819 [i.8], clause 4.6.6

Threat description	As specified in SPS clause 4.7, OAuth HTTP authentication scheme is optional. However, as per [IETF RFC 2616] it relies on the Authorization and WWW-Authenticate headers to distinguish authenticated content so that it can be protected. Proxies and caches, in particular, may fail to adequately protect requests not using these headers.
Threat type	Information disclosure
Vulnerability	Non authenticated, weakly or incorrectly authenticated communication channels
Impact type	Confidentiality
Affected Asset	ASSET-C-01, ASSET-C-02, ASSET-C-07, ASSET-C-08, ASSET-C-09, ASSET-C-10, ASSET-C-11, ASSET-C-32, ASSET-C-34, ASSET-C-36

Threat ID	T-OAuth-G-24
Threat title	Replay of authorized resource server requests, IETF RFC 6819 [i.8], clause 4.6.2
Threat description	An attacker may attempt to replay valid requests to obtain or modify/delete the O-RAN application data.
Threat type	Tampering
Vulnerability	Inadequacy of request validation of Resource server
Impact type	Integrity
Affected Asset	ASSET-C-01, ASSET-C-02, ASSET-C-07, ASSET-C-08, ASSET-C-09, ASSET-C-10, ASSET-C-11, ASSET-C-32, ASSET-C-34, ASSET-C-36

6.2 Authorization Server related Threats

Threat ID	T-OAuth-AS-01
Threat title	Password phishing by counterfeit authorization server, IETF RFC 6819 [i.8], clause 4.2.1
Threat description	A malicious actor may steal confidential data (e.g., Password) by intercepting the O-RAN Application's request and exploit the URL endpoints managed by authorization server.
Threat type	Information disclosure
Vulnerability	Non authenticated, weakly or incorrectly authenticated communication channels
Impact type	Authenticity
Affected Asset	OAuth authorization server

Threat ID	T-OAuth-AS-02
Threat title	Obtaining access tokens from authorization server database, IETF RFC 6819 [i.8], clause 4.3.2
Threat description	A malicious actor may obtain access tokens from the authorization server's database by gaining access to the database or launching a SQL injection attack.
Threat type	Elevation of privilege
Vulnerability	Unsecured database, Insufficient protection for credentials

Impact type	Authorization
Affected Asset	OAuth authorization server

Threat ID	T-OAuth-AS-03
Threat title	Obtaining client secret from authorization server database, IETF RFC 6819 [i.8], clause 4.3.4
Threat description	A malicious actor may obtain valid client_id/secrets combinations from the authorization server's database by gaining access to the database or launching a SQL injection attack.
Threat type	Elevation of privilege
Vulnerability	Unsecured database
Impact type	Authorization
Affected Asset	OAuth authorization server

Threat ID	T-OAuth-AS-04
Threat title	Online guessing of authorization codes, IETF RFC 6819 [i.8], clause 4.4.1.3
Threat description	A malicious actor may try to guess valid authorization code values and send the guessed code value using the grant type code in order to obtain a valid access token.
Threat type	Information disclosure
Vulnerability	Improper authorization process and lack of re-validating the authorization requests
Impact type	Confidentiality
Affected Asset	OAuth authorization server

Threat ID	T-OAuth-AS-05
Threat title	DoS attacks that exhaust resources, IETF RFC 6819 [i.8], clause 4.4.1.11
Threat description	A malicious actor may exhaust the pool of authorization codes of the authorization server by repeatedly directing the O-RAN Application to request the access tokens.
Threat type	Denial of Service
Vulnerability	Improper authorization process and lack of re-validating the authorization requests
Impact type	Availability
Affected Asset	OAuth authorization server

Threat ID	T-OAuth-AS-06
Threat title	CSRF attack against redirect-uri, IETF RFC 6819 [i.8], clause 4.4.1.8
Threat description	CSRF attacks on OAuth approvals may allow a malicious actor to obtain authorization to OAuth protected resources without the consent of the O-RAN Applications.
Threat type	Elevation of privilege
Vulnerability	Improper authorization process and lack of re-validating the authorization requests

Impact type	Authorization
Affected Asset	OAuth authorization server

Threat ID	T-OAuth-AS-07
Threat title	Resource owner impersonation, IETF RFC 6819 [i.8], clause 4.4.1.10
Threat description	A malicious actor could obtain unauthorized access to a resource by using an existing authenticated session that causes the OAuth 2.0 authorization server to improperly bypass resource owner consent.
Threat type	Elevation of privilege
Vulnerability	Improper authorization process and lack of re-validating the authorization requests
Impact type	Authorization
Affected Asset	OAuth authorization server

Threat ID	T-OAuth-AS-08
Threat title	DoS using manufactured authorization "codes", IETF RFC 6819 [i.8], clause 4.4.1.12
Threat description	An attacker who owns a botnet may locate the redirect URIs of client that listen on HTTP, access them with random authorization "codes", and cause a large number of HTTPS connections to be concentrated onto the authorization server. This can result in a denial-of-service (DoS) attack on the authorization server.
Threat type	Denial of Service
Vulnerability	Non authenticated, weakly or incorrectly authenticated communication channels
Impact type	Availability
Affected Asset	OAuth authorization server

Threat ID	T-OAuth-AS-09
Threat title	Obtaining user passwords from authorization server database, IETF RFC 6819 [i.8], clause 4.4.3.5
Threat description	An attacker may obtain valid username/password combinations from the authorization server's database by gaining access to the database or launching a SQL injection attack.
Threat type	Information disclosure
Vulnerability	Unsecured database, Insufficient protection for credentials
Impact type	Authorization Information disclosure
Affected Asset	OAuth authorization server

Threat ID	T-OAuth-AS-10
Threat title	Obtaining the resource owner password credentials by online guessing, IETF RFC 6819 [i.8], clause 4.4.3.6
Threat description	An attacker may try to guess valid username/password combinations using the grant type "password".

Threat type	Information disclosure
Vulnerability	Unsecure credentials
Impact type	Authorization
Affected Asset	OAuth authorization server

Threat ID	T-OAuth-AS-11
Threat title	Obtaining refresh token from authorization server database, IETF RFC 6819 [i.8], clause 4.5.2
Threat description	This threat is applicable if the authorization server stores refresh tokens as handles in a database. An attacker may obtain refresh tokens from the authorization server's database by gaining access to the database or launching a SQL injection attack.
Threat type	Information disclosure
Vulnerability	Unsecured database
Impact type	Authorization
Affected Asset	OAuth authorization server

7 Risk Assessment

7.1 Security Controls

The following security controls should be considered for the security threat and risk analysis:

- Control-1: OAuth 2.0
- Control-2: IAM (using RBAC)
- Control-3: Certificate Management
- Control-4: Encryption for Data at Rest
- Control-5: Encryption for Data in Motion (eg., TLS)

7.2 OAuth 2.0 Risk Analysis

This clause provides risk assessment tables for each of the identified assets. These tables list the assets, threats, impacts, and possible security controls. The following risk assessment matrix is used to assess the risk score as defined in [i.20].

Table 7-2-1 OAuth 2.0 risk analysis

Asset-Id	Asset Name	Threat-Id	Threat Description (Brief)	Impact/ Likelihood Raw Score	Possible Security Controls	Security Control-id
ASSET-C-01	SMO	T-OAuth-G-01	Obtaining client secrets	Impact = High Likelihood = Medium	Data encryption, OAuth 2.0	Control-1 Control-2 Control-4 Control-5
ASSET-C-02	Near-RT RIC software					
ASSET-C-07	O-eNB					

ASSET-C-08	O-Cloud					
ASSET-C-09	xApps					
ASSET-C-10	rApps					
ASSET-C-11	Non-RT RIC					
ASSET-C-32	O-RU					
ASSET-C-34	O-DU					
ASSET-C-36	O-CU Host, includes O-CU-CP and O-CU-UP software					
ASSET-C-01	SMO	T-OAuth-G-02	Obtaining refresh tokens	Impact = High Likelihood = Medium	Data encryption	Control-4 Control-5
ASSET-C-02	Near-RT RIC software					
ASSET-C-07	O-eNB					
ASSET-C-08	O-Cloud					
ASSET-C-09	xApps					
ASSET-C-10	rApps					
ASSET-C-11	Non-RT RIC					
ASSET-C-32	O-RU					
ASSET-C-34	O-DU					
ASSET-C-36	O-CU Host, includes O-CU-CP and O-CU-UP software					
ASSET-C-01	SMO	T-OAuth-G-03	Obtaining access tokens	Impact = High	Data encryption, IAM	Control-2 Control-4

ASSET-C-02	Near-RT RIC software			Likelihood = Medium		
ASSET-C-07	O-eNB					
ASSET-C-08	O-Cloud					
ASSET-C-09	xApps					
ASSET-C-10	rApps					
ASSET-C-11	Non-RT RIC					
ASSET-C-32	O-RU					
ASSET-C-34	O-DU					
ASSET-C-36	O-CU Host, includes O-CU-CP and O-CU-UP software					
ASSET-C-01	SMO	T-OAuth-G-04	Eavesdropping access tokens	Impact = High Likelihood = Medium	Authentication Data encryption	Control-5
ASSET-C-02	Near-RT RIC software					
ASSET-C-07	O-eNB					
ASSET-C-08	O-Cloud					
ASSET-C-09	xApps					
ASSET-C-10	rApps					
ASSET-C-11	Non-RT RIC					
ASSET-C-32	O-RU					
ASSET-C-34	O-DU					
ASSET-C-36	O-CU Host, includes O-CU-CP and O-CU-UP software					

ASSET-C-01	SMO	T-OAuth-G-05	Disclosure of client credentials during transmission	Impact = High Likelihood = Medium	Authentication Data encryption	Control-5
ASSET-C-02	Near-RT RIC software					
ASSET-C-07	O-eNB					
ASSET-C-08	O-Cloud					
ASSET-C-09	xApps					
ASSET-C-10	rApps					
ASSET-C-11	Non-RT RIC					
ASSET-C-32	O-RU					
ASSET-C-34	O-DU					
ASSET-C-36	O-CU Host, includes O-CU-CP and O-CU-UP software					
ASSET-C-01	SMO	T-OAuth-G-06	Open redirectors on client	Impact = High Likelihood = Medium	Authentication, IAM	Control-1, Control-5
ASSET-C-02	Near-RT RIC software					
ASSET-C-07	O-eNB					
ASSET-C-08	O-Cloud					
ASSET-C-09	xApps					
ASSET-C-10	rApps					
ASSET-C-11	Non-RT RIC					
ASSET-C-32	O-RU					
ASSET-C-34	O-DU					
ASSET-C-36	O-CU Host, includes O-CU-CP and O-					

	CU-UP software					
ASSET-C-01	SMO	T-OAuth-G-07	Obtain a deployment-specific secret	Impact = High Likelihood = Medium	Data encryption, Access control	Control-2 Control-4
ASSET-C-02	Near-RT RIC software					
ASSET-C-07	O-eNB					
ASSET-C-08	O-Cloud					
ASSET-C-09	xApps					
ASSET-C-10	rApps					
ASSET-C-11	Non-RT RIC					
ASSET-C-32	O-RU					
ASSET-C-34	O-DU					
ASSET-C-36	O-CU Host, includes O-CU-CP and O-CU-UP software					
ASSET-C-01	SMO	T-OAuth-G-08	Obtaining authorization by code substitution (OAuth Login)	Impact = High Likelihood = Medium	Identity Management	Control-1 Control-2
ASSET-C-02	Near-RT RIC software					
ASSET-C-07	O-eNB					
ASSET-C-08	O-Cloud					
ASSET-C-09	xApps					
ASSET-C-10	rApps					
ASSET-C-11	Non-RT RIC					
ASSET-C-32	O-RU					
ASSET-C-34	O-DU					

ASSET-C-36	O-CU Host, includes O-CU-CP and O-CU-UP software					
ASSET-C-01	SMO	T-OAuth-G-09	Access token leak in transport/endpoints	Impact = High Likelihood = Medium	Authentication Data encryption	Control-5
ASSET-C-02	Near-RT RIC software					
ASSET-C-07	O-eNB					
ASSET-C-08	O-Cloud					
ASSET-C-09	xApps					
ASSET-C-10	rApps					
ASSET-C-11	Non-RT RIC					
ASSET-C-32	O-RU					
ASSET-C-34	O-DU					
ASSET-C-36	O-CU Host, includes O-CU-CP and O-CU-UP software					
ASSET-C-01	SMO	T-OAuth-G-10	Malicious client obtains existing authorization by fraud	Impact = High Likelihood = Medium	Access control	Control-2
ASSET-C-02	Near-RT RIC software					
ASSET-C-07	O-eNB					
ASSET-C-08	O-Cloud					
ASSET-C-09	xApps					
ASSET-C-10	rApps					
ASSET-C-11	Non-RT RIC					
ASSET-C-32	O-RU					

ASSET-C-34	O-DU					
ASSET-C-36	O-CU Host, includes O-CU-CP and O-CU-UP software					
ASSET-C-01	SMO					
ASSET-C-02	Near-RT RIC software	T-OAuth-G-11	Malicious client obtains authorization	Impact = High Likelihood = Medium	Identity Management, Data Encryption	Control-1 Control-2 Control-4 Control-5
ASSET-C-07	O-eNB					
ASSET-C-08	O-Cloud					
ASSET-C-09	xApps					
ASSET-C-10	rApps					
ASSET-C-11	Non-RT RIC					
ASSET-C-32	O-RU					
ASSET-C-34	O-DU					
ASSET-C-36	O-CU Host, includes O-CU-CP and O-CU-UP software					
ASSET-C-01	SMO	T-OAuth-G-12	Accidental exposure of passwords at client site	Impact = High Likelihood = Medium	Data Encryption	Control-4 Control-5
ASSET-C-02	Near-RT RIC software					
ASSET-C-07	O-eNB					
ASSET-C-08	O-Cloud					
ASSET-C-09	xApps					
ASSET-C-10	rApps					
ASSET-C-11	Non-RT RIC					

ASSET-C-32	O-RU					
ASSET-C-34	O-DU					
ASSET-C-36	O-CU Host, includes O-CU-CP and O-CU-UP software					
ASSET-C-01	SMO	T-OAuth-G-13	Client obtains scopes without end-user authorization and possible misconfigurations	Impact = High Likelihood = Medium	Access control, Identity Management	Control-1 Control-2
ASSET-C-02	Near-RT RIC software					
ASSET-C-07	O-eNB					
ASSET-C-08	O-Cloud					
ASSET-C-09	xApps					
ASSET-C-10	rApps					
ASSET-C-11	Non-RT RIC					
ASSET-C-32	O-RU					
ASSET-C-34	O-DU					
ASSET-C-36	O-CU Host, includes O-CU-CP and O-CU-UP software					
ASSET-C-01	SMO	T-OAuth-G-14	Client obtains refresh token through automatic authorization	Impact = High Likelihood = Medium	Access control, Identity Management	Control-1 Control-2
ASSET-C-02	Near-RT RIC software					
ASSET-C-07	O-eNB					
ASSET-C-08	O-Cloud					
ASSET-C-09	xApps					
ASSET-C-10	rApps					

ASSET-C-11	Non-RT RIC					
ASSET-C-32	O-RU					
ASSET-C-34	O-DU					
ASSET-C-36	O-CU Host, includes O-CU-CP and O-CU-UP software					
ASSET-C-01	SMO	T-OAuth-G-15	Eavesdropping refresh tokens from authorization server	Impact = High Likelihood = Medium	Authentication Data encryption	Control-5
ASSET-C-02	Near-RT RIC software					
ASSET-C-07	O-eNB					
ASSET-C-08	O-Cloud					
ASSET-C-09	xApps					
ASSET-C-10	rApps					
ASSET-C-11	Non-RT RIC					
ASSET-C-32	O-RU					
ASSET-C-34	O-DU					
ASSET-C-36	O-CU Host, includes O-CU-CP and O-CU-UP software					
ASSET-C-01	SMO	T-OAuth-G-16	Obtaining refresh token by online guessing	Impact = High Likelihood = Medium	Access control, Identity Management	Control-1 Control-2
ASSET-C-02	Near-RT RIC software					
ASSET-C-07	O-eNB					
ASSET-C-08	O-Cloud					
ASSET-C-09	xApps					

ASSET-C-10	rApps					
ASSET-C-11	Non-RT RIC					
ASSET-C-32	O-RU					
ASSET-C-34	O-DU					
ASSET-C-36	O-CU Host, includes O-CU-CP and O-CU-UP software					
ASSET-C-01	SMO	T-OAuth-G-17	Refresh token phishing by counterfeit authorization server	Impact = High Likelihood = Medium	Authentication, Data encryption	Control-5
ASSET-C-02	Near-RT RIC software					
ASSET-C-07	O-eNB					
ASSET-C-08	O-Cloud					
ASSET-C-09	xApps					
ASSET-C-10	rApps					
ASSET-C-11	Non-RT RIC					
ASSET-C-32	O-RU					
ASSET-C-34	O-DU					
ASSET-C-36	O-CU Host, includes O-CU-CP and O-CU-UP software					
ASSET-C-01	SMO	T-OAuth-G-18	Guessing access tokens	Impact = High Likelihood = Medium	Access control, Identity Management	Control-1 Control-2
ASSET-C-02	Near-RT RIC software					
ASSET-C-07	O-eNB					

ASSET-C-08	O-Cloud					
ASSET-C-09	xApps					
ASSET-C-10	rApps					
ASSET-C-11	Non-RT RIC					
ASSET-C-32	O-RU					
ASSET-C-34	O-DU					
ASSET-C-36	O-CU Host, includes O-CU-CP and O-CU-UP software					
ASSET-C-01	SMO	T-OAuth-G-19	Access token phishing by counterfeit resource server	Impact = High Likelihood = Medium	Authentication, Data encryption	Control-5
ASSET-C-02	Near-RT RIC software					
ASSET-C-07	O-eNB					
ASSET-C-08	O-Cloud					
ASSET-C-09	xApps					
ASSET-C-10	rApps					
ASSET-C-11	Non-RT RIC					
ASSET-C-32	O-RU					
ASSET-C-34	O-DU					
ASSET-C-36	O-CU Host, includes O-CU-CP and O-CU-UP software					
ASSET-C-01	SMO	T-OAuth-G-20	Abuse of token by legitimate resource server or client	Impact = High Likelihood = Medium	Access control, Identity Management	Control-1 Control-2
ASSET-C-02	Near-RT RIC software					

ASSET-C-07	O-eNB					
ASSET-C-08	O-Cloud					
ASSET-C-09	xApps					
ASSET-C-10	rApps					
ASSET-C-11	Non-RT RIC					
ASSET-C-32	O-RU					
ASSET-C-34	O-DU					
ASSET-C-36	O-CU Host, includes O-CU-CP and O-CU-UP software					
ASSET-C-01	SMO	T-OAuth-G-21	Token leakage via log files and HTTP referrers	Impact = High Likelihood = Medium	Access control, Identity Management, Data encryption	Control-1 Control-2 Control-5
ASSET-C-02	Near-RT RIC software					
ASSET-C-07	O-eNB					
ASSET-C-08	O-Cloud					
ASSET-C-09	xApps					
ASSET-C-10	rApps					
ASSET-C-11	Non-RT RIC					
ASSET-C-32	O-RU					
ASSET-C-34	O-DU					
ASSET-C-36	O-CU Host, includes O-CU-CP and O-CU-UP software					
ASSET-C-01	SMO	T-OAuth-G-22	User unintentionally grants too much access scope	Impact = High	Access control, Identity Management	

ASSET-C-02	Near-RT RIC software			Likelihood = Medium		Control-1 Control-2
ASSET-C-07	O-eNB					
ASSET-C-08	O-Cloud					
ASSET-C-09	xApps					
ASSET-C-10	rApps					
ASSET-C-11	Non-RT RIC					
ASSET-C-32	O-RU					
ASSET-C-34	O-DU					
ASSET-C-36	O-CU Host, includes O-CU-CP and O-CU-UP software					
ASSET-C-01	SMO	T-OAuth-G-23	Leak of confidential data in HTTP proxies	Impact = High Likelihood = Medium	Authentication, Data encryption	Control-5
ASSET-C-02	Near-RT RIC software					
ASSET-C-07	O-eNB					
ASSET-C-08	O-Cloud					
ASSET-C-09	xApps					
ASSET-C-10	rApps					
ASSET-C-11	Non-RT RIC					
ASSET-C-32	O-RU					
ASSET-C-34	O-DU					
ASSET-C-36	O-CU Host, includes O-CU-CP and O-CU-UP software					

ASSET-C-01	SMO	T-OAuth-G-24	Replay of authorized resource server requests	Impact = High Likelihood = Medium	Authentication, Data encryption, Identity Management	Control-2 Control-5
ASSET-C-02	Near-RT RIC software					
ASSET-C-07	O-eNB					
ASSET-C-08	O-Cloud					
ASSET-C-09	xApps					
ASSET-C-10	rApps					
ASSET-C-11	Non-RT RIC					
ASSET-C-32	O-RU					
ASSET-C-34	O-DU					
ASSET-C-36	O-CU Host, includes O-CU-CP and O-CU-UP software					

8 Void

The following security controls should be considered for the security threat and risk analysis:

- Control-1: OAuth 2.0
- Control-2: IAM (using RBAC)
- Control-3: Certificate Management
- Control-4: Encryption for Data at Rest
- Control-5: Encryption for Data in Motion

9 OAuth 2.0 Authorization framework

9.1 Overview

Figure 9.1-1 depicts the basic OAuth 2.0 framework for O-RAN architecture elements and applications. This general authorization framework is generalized from clause 1 of IETF RFC 6749 [i.5].

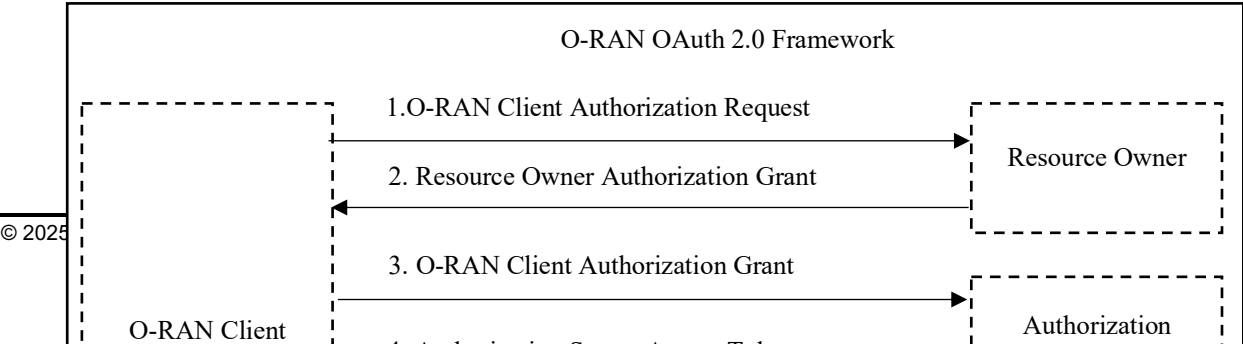


Figure 9.1-1 OAuth 2.0 framework for O-RAN architecture elements and applications

O-RAN OAuth 2.0 framework defines four roles from clause 1 of IETF RFC 6749 [i.5]:

- O-RAN client: An O-RAN client is an O-RAN architecture element or application like an rApp or xApp acting as an OAuth client.
- Authorization server: The server issuing access tokens to the O-RAN client after successfully authenticating the resource owner and obtaining authorization.
- Resource owner: “An entity capable of granting access to a protected resource. When the resource owner is a person, it is referred to as an end-user.” ([i.5], clause 1.1)
- Resource server: “The server hosting the protected resources, capable of accepting and responding to protected resource requests using access tokens.” ([i.5], clause 1.1)

9.2 Key Issues

9.2.1 Key Issue #1: Registration of O-RAN Architecture Elements in OAuth Authorization Server

9.2.1.1 Key issue detail

This key issue addresses the security threats during the registration of O-RAN client (e.g., xApp, rApp) in the authorization server.

9.2.1.2 Security threats

The threats related to the above key issue are T-OAuth-G-05, T-OAuth-G-06, T-OAuth-G-12 as defined in clause 6.1.

9.2.2 Key Issue #2: Provisioning of Client ID and secrets to O-RAN Architecture Elements

9.2.2.1 Key issue detail

This key issue addresses the security threats while provisioning the client ID and secrets for O-RAN client (e.g., xApp, rApp).

9.2.2.2 Security threats

The threats related to the above key issue are T-OAuth-G-01, T-OAuth-G-03, T-OAuth-G-04, T-OAuth-G-05, T-OAuth-G-09, T-OAuth-G-18, T-OAuth-G-19, T-OAuth-G-20, T-OAuth-G-21, T-OAuth-AS-10 as defined in clause 6.1 and clause 6.2.

9.2.3 Key Issue #3: Provisioning of Authorization Grants for O-RAN Architecture Elements

9.2.3.1 Key issue detail

This key issue addresses the security threats while provisioning the authorization grants for O-RAN client (e.g., xApp, rApp).

9.2.3.2 Security threats

The threats related to the above key issue are T-OAuth-G-08, T-OAuth-G-10, T-OAuth-G-11, T-OAuth-G-13 as defined in clause 6.1.

9.2.4 Key Issue #4: Revocation of Client ID and secrets for O-RAN Architecture Elements

9.2.4.1 Key issue detail

This key issue addresses the security threats related to revocation of client ID and secrets for O-RAN client (e.g., xApp, rApp).

9.2.4.2 Security threats

The threats related to the above key issue are T-OAuth-G-02, T-OAuth-G-14, T-OAuth-G-15, T-OAuth-G-17, T-OAuth-AS-11 as defined in clause 6.1 and clause 6.2.

9.2.5 Key Issue #5: Revocation of Authorization Grants for O-RAN Architecture Elements

9.2.5.1 Key issue detail

This key issue addresses the security threats related to revocation of authorization grants for O-RAN client (e.g., xApp, rApp).

9.2.5.2 Security threats

The threats related to the above key issue are T-OAuth-G-10, T-OAuth-AS-04, T-OAuth-AS-07 as defined in clause 6.1 and clause 6.2.

9.2.6 Key Issue #6: Secure transfer of client secrets to clients

9.2.6.1 Key issue detail

The need is to identify,
How are the client secrets transferred to potential clients?
Whether the client secrets will be known at development time or runtime?

9.2.6.2 Security threats

The threats related to the above key issue are T-OAuth-G-01, T-OAuth-G-05, T-OAuth-G-12, T-OAuth-AS-01, T-OAuth-AS-03 as defined in clause 6.1 and clause 6.2.

9.2.7 Key Issue #7: Secure transit and storage of access and refresh tokens

9.2.7.1 Key issue detail

It needs to be identified what possible mechanisms exists to ensure the integrity and confidentiality of access and refresh tokens.

9.2.7.2 Security threats

The threats related to the above key issue are T-OAuth-G-02, T-OAuth-G-03, T-OAuth-G-04, T-OAuth-G-20, T-OAuth-G-15, T-OAuth-G-16, T-OAuth-G-18 as defined in clause 6.1.

9.2.8 Key Issue #8: Defining expiration times for access tokens and client secrets

9.2.8.1 Key issue detail

It needs to be identified if

- Access tokens need to be configured with the minimal expiration time and in what use cases
- Client secrets need to be revoked and refreshed regularly and if so at what intervals

9.2.8.2 Security threats

The threats related to the above key issue are T-OAuth-G-10, T-OAuth-AS-01, T-OAuth-AS-02, T-OAuth-AS-03 as defined in clause 6.1 and clause 6.2.

9.2.9 Key Issue #9: Consumers based on Client_IDs created with valid administration

9.2.9.1 Key issue detail

It needs to be identified if

- Consumers who connect are only those validated by administrators
- Dynamic clients are possible or not

9.2.9.2 Security threats

The threat related to the above key issue is T-OAuth-G-11 as defined in clause 6.1.

9.2.10 Key Issue #10: Scope creation

9.2.10.1 Key issue detail

It needs to be checked as to how scopes are designed, provisioned and enforced. Key topics are

- How scopes are designed by the resource servers including the clearance and access levels
- How scopes are transferred the authorization server
- How scopes are requested by the client
- How scopes are created in the authorization server

9.2.10.2 Security threats

The threats related to the above key issue are T-OAuth-G-11, T-OAuth-G-13 as defined in clause 6.1.

9.2.11 Key Issue #11: Need to support RBAC for the O-RAN resource servers

The O-RAN resource servers need to be accessed by only those users who are authorized to carry out the specific task. Hence, we need a mechanism to design and map the O-RAN resource server to the users for access control.

9.2.11.1 Key issue detail

It needs to be checked as to

- How to limit the access of resources to specified users (least privilege) and only for the intended task.
- How to manage access (assign users to permissions related to the resource servers) in a fast and efficient manner.
- How to retain a role only for the period of time required and no longer.

9.2.11.2 Security threats

The threats related to limited access control are below.

- Malicious attacks with open access to the entire system of assets and data
- Insider threats

- Privilege stagnation over a period of time

Editor notes: Some of the threats id's such as T-O-RAN-03,T-O-RAN-06,T-R1-01,T-R1-05,T-AppLCM-04,T-GEN-02,T-SMO-10 in clause 5.4 of the present document are mapped and identified in the O-RAN Threat Modelling and Remediation Analysis document [i.20], need to be discussed further.

9.2.12 Key Issue #12: Need to support secure transmission

The key issues address the need to support for secure transmission of request and response between client and authorization server, between client and resource server.

The O-RAN related HTTP client (e.g.: R1, A1, O2) need to make a request to the authorization server and receive a response with respect to the access token. After that the access token has to be transferred from the client to the resource server.

9.2.12.1 Key issue detail

It needs to be checked as to

- How the client will send request details including credentials related to client_id and client_secret to the authorization server
- How the client will receive the response from the authorization server
- How the client will send the access token to the resource server

9.2.12.2 Security threats

The threats related to insecure transmission of request and response are T-OAuth-G-01, T-OAuth-G-03, T-OAuth-G-05 as defined in clause 6.1.

9.3 Solutions

9.3.1 Solution #1: Registration of rApp

9.3.1.1 Background

This use case allows an rApp to register to the SMO as specified in R1GAP [i.13] [i.14]. The registration procedure is specified as part of Service Management and Exposure services in R1GAP **Error! Reference source not found.** [i.14].

9.3.1.2 Solution Description

Table 9.3.1-1 rApp Registration use case

Use Case Stage	Evolution / Specification	<<Uses>> Related use
Goal	1) Register the rApp to the SME services Producer in SMO framework.	
Actors and Roles	- rApp in the role of Service Producer and/or Service Consumer - SME in the role of SME services Producer.	
Assumptions	n/a	
Preconditions	The rApp is instantiated.	
Begins when	The rApp Management SMOS is ready for registration.	
Step 1 (M)	The rApp Management SMOS sends a registration request to the SME in the SMO passing relevant information needed.	
Step 1 (M)	The SME process the rApp registration request that may include. - performing authentication, - if rApp is authenticated, rAppId will be assigned to rApp.	
Step 3 (M)	Response to the registration request along with the rAppId.	
Ends when	The rApp is allocated with an rAppId.	
Exceptions	n/a	
Post Conditions	The rApp is registered with the SMO framework and an rAppId is allocated to the rApp.	
Traceability	REQ-R1-SME-rAppreg-FUN1, REQ-R1-SME-rAppreg-FUN2.	

```

@startuml
    skinparam ParticipantPadding 5
    skinparam BoxPadding 10
    skinparam defaultFontSize 12
    skinparam lifelineStrategy solid
    autonumber
    box " SMO " #cadetBlue
    participant "rApp Management SMOS" as rApp_M

    participant "SME(SMOS)" as SME
    rApp_M -> SME: App Manager sends a registration request \n to the SME service based on endpoint details \n and passing relevant information needed
    SME->SME :
    note right
    Registration processing
    assign rAppid using
    Authorization service
    end note
    SME -> rApp_M :Response to the registration request (rAppid)

    Note Over rApp_M
    rApp Management assign
    rAppid to rApp as per
    App LCM process
    end note
    endbox
@enduml

```

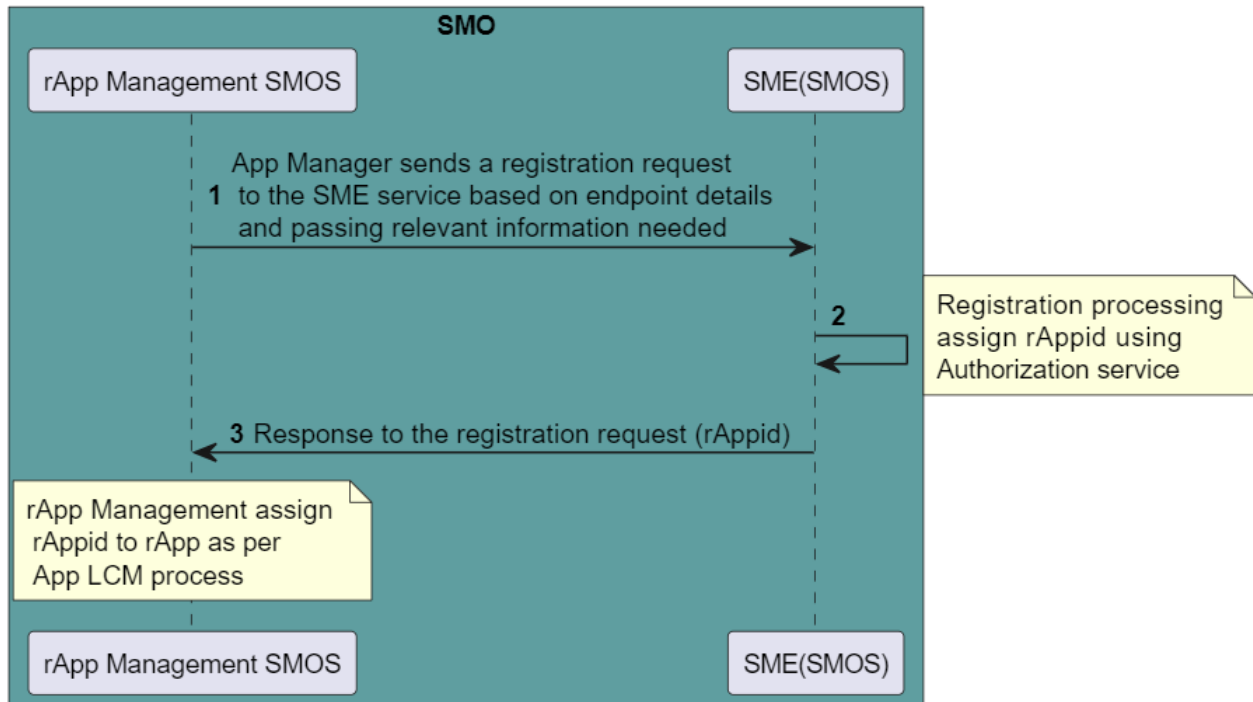


Figure 9.3.1-1 rApp Registration use case sequence diagram

9.3.1.3 Impacts on Existing O-RAN Architecture Elements and Interfaces

R1 interface, rApp and SME are the O-RAN architecture elements and interfaces involved in the proposed use cases as specified in Decoupled SMO [i.15] [i.16].

1) Service Management and Exposure (SME) SMOS

The SME service is introduced in the [i.8] clause 3.1. While the SME was initially targeted for management and exposure of rApp services in the [i.8], its services are applicable to any entities within the SMO that fulfil the roles of Service Producer and/or Service Consumer.

In a decomposed service-based SMO architecture, the SME can be leveraged as a generic SMOS, handling service management and exposure for any SMOSs within SMO. The SME SMOS provides a set of common SMO capabilities that are needed for all SMO Services.

Several of the SME services are applicable to any SMOSs:

- SMOS registration,
- SMOS discovery,
- Authentication and authorization to access a SMOS,
- Communication support between SMOS Service Producers and SMOS Service Consumers (SMOS Communication),
- Bootstrap of new SMOSs (optional),
- Heartbeat of SMOSs (optional).

2) rApp Management (SMOS)

The rApp management service handles management aspects specific to rApps such as:

- rApp Configuration Management (CM): enables management of the rApp configuration settings,
- rApp Performance Management (PM): offers support for performance reporting on an rApp performance, or of the rApp communication with the non-RT RIC framework (rApp usage of R1 interface),
- rApp Fault Management (FM): provides the fault reporting capabilities to rApps, so they can consume it to report faults that are related to the rApp, or to the rApp communication with the non-RT RIC framework (rApp usage of R1 interface),
- rApp Logging Information: enables rApps to convey their logging information to the non-RT RIC framework,
- rApp lifecycle management services, that expose services that allow to trigger rApp software lifecycle management.

9.3.1.4 Solution Evaluation

Key Issue #1: Registration of O-RAN architecture elements in OAuth authorization server

9.3.1.5 Potential Security Recommendations

- OAuth protection of the R1 interface is already mandatory in the O-RAN Security Requirements and Controls Specifications [i.2] SEC-CTL-R1-3.

9.3.2 Solution #2: Request access token for R1 services

9.3.2.1 Background

This use case allows an rApp to request an access token for access to R1 services as specified in R1GAP [i.13] [i.14]. The request access procedure is specified as part of Service Management and Exposure services in R1GAP **Error! Reference source not found.** [i.14].

An rApp in the role of Service Consumer will be able to request the token and once the token is available, the rApp in the role of Service Consumer will be able access the R1 Services for which it has been authorized.

9.3.2.2 Solution Description

Table 9.3.2-1 Request access token for R1 services use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	To enable rApp to request an access token for access to R1 services.	
Actors and Roles	rApp in the role of Service Consumer that requests token for access to R1 services. SME as SMOS in the role of SME services Producer support authorization service	
Assumptions	rAppId and credentials have been provisioned.	
Pre-conditions	R1 interface is established	
Begins when	rApp has determined need for an access token to access R1 services and resources.	
Step 1 (M)	rApp sends Access token request containing the rAppId and optionally token scope.	SPS [i.1], clause 4.7.2.3
Step 2 (M)	SME authenticate the rApp and if the rApp token request is authorized, then an access token is generated.	SPS [i.1], clause 4.7.2.3
Step 3 (M)	SME respond with access token.	SPS [i.1], clause 4.7.2.3
Ends when	rApp has received an access token for access to R1 services.	
Exceptions	n/a	
Post-conditions	The rApp is authorized for accessing R1 services.	
Traceability	REQ-R1-A-FUN1	

```

@startuml
    skinparam ParticipantPadding 5
    skinparam BoxPadding 10
    skinparam defaultFontSize 12
    skinparam lifelineStrategy solid
    autonumber
    box "non-RT RIC" #whitesmoke
        box #ivory
            participant rApp as rApp
        endbox
        box "SMO" #cadetBlue
            participant "SME (SMOS)" as SME
        endbox
        rApp -> SME: <<R1>> Access token request (rAppId)
        SME->SME:
        note right
            Check Authorization
            (Authenticate the rApp and
            check if authorized, if
            rApp is authorized,
            generate an access token)
        end note
        SME -> rApp :<<R1>> Access token response (access token)
    @enduml

```

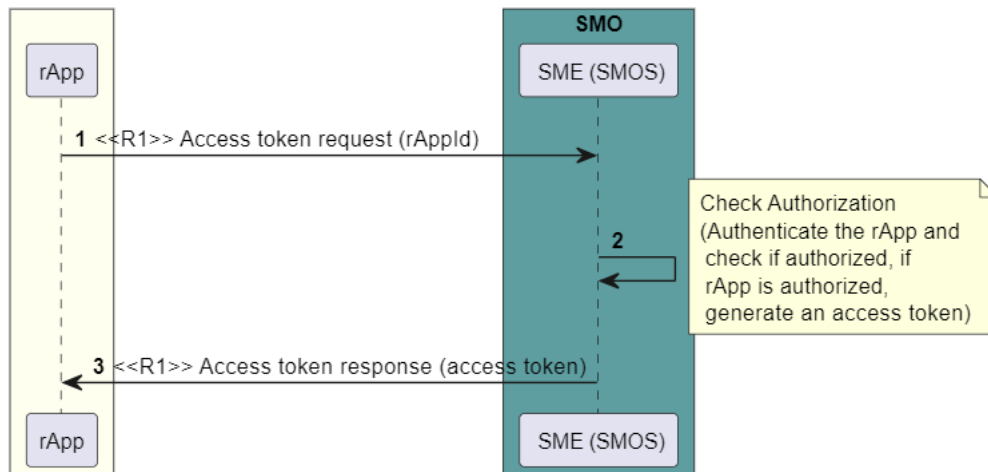


Figure 9.3.2-1 Request access token for R1 services use case sequence diagram

9.3.2.3 Impacts on Existing O-RAN Architecture Elements and Interfaces

R1 interface, rApp and SME are the O-RAN architecture elements and interfaces involved in the proposed use cases as specified in Decoupled SMO [i.15] [i.16].

1) Service Management and Exposure (SME) SMOS

The SME service is introduced in the [i.8] clause 3.1. While the SME was initially targeted for management and exposure of rApp services in the [i.8], its services are applicable to any entities within the SMO that fulfil the roles of Service Producer and/or Service Consumer.

In a decomposed service-based SMO architecture, the SME can be leveraged as a generic SMOS, handling service management and exposure for any SMOSs within SMO. The SME SMOS provides a set of common SMO capabilities that are needed for all SMO Services. Several of the SME services are applicable to any SMOSs:

- SMOS registration,
- SMOS discovery,
- Authentication and authorization to access a SMOS,
- Communication support between SMOS Service Producers and SMOS Service Consumers (SMOS Communication),
- Bootstrap of new SMOSs (optional),
- Heartbeat of SMOSs (optional).

2) SME functions in the role of SME services Producer

- supports the functionality of OAuth 2.0 authorization server,
- support authentication and authorization of an rApp to determine which R1 services an rApp can access,
- provide response of success or failure to access token request.

3) rApp

- supports initiating the procedure to request access token,
- supports initiating service request with the available access token.

9.3.2.4 Solution Evaluation

Key Issue #3: Provisioning of Authorization Grants for R1 services

9.3.2.5 Potential Security Recommendation

- Use OAuth 2.0, as specified in O-RAN Security Protocols Specifications [i.1], clause 4.7 to implement authorization in the R1 interface.
- Enable rApp to request OAuth 2.0 access tokens for R1 services.
- Enable the SME to act as both an OAuth 2.0 resource and an OAuth 2.0 authorization server.

9.3.3 Solution #3: R1 services request with access token

9.3.3.1 Background

This use case discusses R1 service Producer to verify that rApp is authorized to access the R1 service as specified in R1GAP [i.13] [i.14]. The service request procedure is specified as part of Service Management and Exposure services in R1GAP **Error! Reference source not found.**[i.14].

9.3.3.2 Solution Description

Table 9.3.3-1 R1 Service request using access token use case

Use case stage	Evolution / specification	<<Uses>> Related use
Goal	To enable R1 service Producer to verify that rApp is authorized to access the R1 service.	
Actors and Roles	rApp in the role of Service Consumer that requests access to an R1 service with access token. R1 service producer in the role of SME services Producer that supports the validation of the access token and provides access to the requested service.	
Assumptions	rApp has requested and received an access token for the R1 service for which service access is requested.	
Pre-conditions	R1 interface is established.	
Begins when	rApp initiates a request for an R1 Service to R1 Service producer.	
Step 1 (M)	rApp sends service request containing an access token to R1 service Producer	
Step 2 (M)	R1 service Producer verifies integrity and claims in the access token. If successful, the requested service is executed.	SPS [i.1] clause 4.7.2.4]
Step 3 & 4 (M)	R1 service Producer sends response to the service request if it has valid token	
Step 5 & 6 (O)	The verification of integrity and claims in step 3 was unsuccessful due to invalid or expired token, the requested service is not executed, and R1 service Producer sends a response with appropriate error code.	
Ends when	R1 service Producer has responded to the service request.	
Post-conditions	-	
Traceability	REQ-R1-SME-A-FUN2	

```

@startuml
    !pragma teoz true
    skinparam ParticipantPadding 5
    skinparam BoxPadding 10
    skinparam defaultFontSize 12
    skinparam lifelineStrategy solid
    autonumber
    box #whitesmoke
        box #ivory
            participant rApp as rApp
        endbox
        box "SMO" #cadetBlue
            participant "SME" as SME
            participant "R1 service Producer m" as SP
        endbox
    endbox
    rApp->>SP: <<R1>> Service request (access token)
    SP->>SP
    note right
        Validate token
        (Verify integrity and claims
        in the access token.
        If successful, execute
        the requested service).
    end note
    alt
        SP->>SP: Valid Token
        SP->>rApp: <<R1>> Service response (success)
        SP->>SP: Invalid or Expired Token
        SP->>rApp: <<R1>> Service response (failure)
    end ref
end ref
@enduml

```

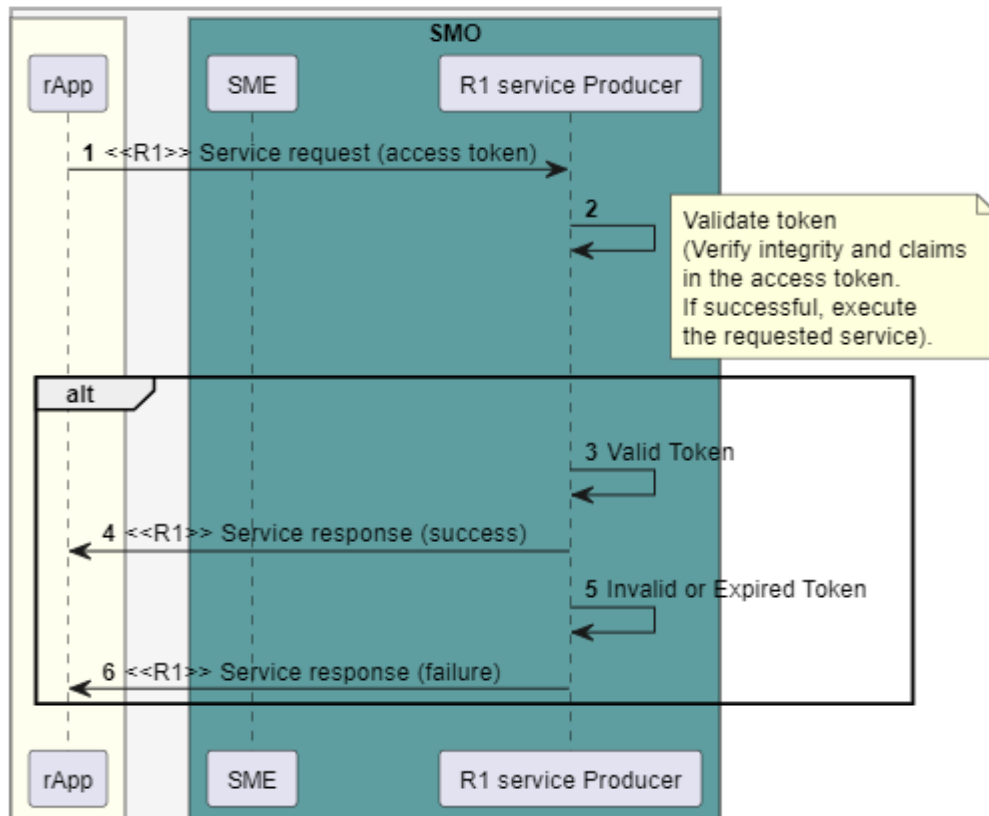


Figure 9.3.3-1 R1 Service request using access token use case sequence diagram

9.3.3.3 Impacts on Existing O-RAN Architecture Elements and Interfaces

R1 interface, rApp and SME are the O-RAN architecture elements and interfaces involved in the proposed use cases as specified in Decoupled SMO [i.15] [i.16].

1) Service Management and Exposure (SME) SMOS

The SME service is introduced in the [i.13] clause 3.1. While the SME was initially targeted for management and exposure of rApp services in the [i.13], its services are applicable to any entities within the SMO that fulfil the roles of Service Producer and/or Service Consumer.

In a decomposed service-based SMO architecture, the SME can be leveraged as a generic SMOS, handling service management and exposure for any SMOSs within SMO. The SME SMOS provides a set of common SMO capabilities that are needed for all SMO Services.

Several of the SME services are applicable to any SMOSs:

- SMOS registration,
- SMOS discovery,
- Authentication and authorization to access a SMOS,
- Communication support between SMOS Service Producers and SMOS Service Consumers (SMOS Communication),
- Bootstrap of new SMOSs (optional),
- Heartbeat of SMOSs (optional).

2) SME in the role of SME services Producer

- supports the functionality of OAuth 2.0 authorization server,
- support authentication and authorization of an rApp to determine which R1 services an rApp can access,
- provide response of success or failure to access token request

3) rApp

- initiating the procedure to request access token,
- initiating service request with the available access token.

4) R1 Service Producer

- provider of an R1 service.

9.3.3.4 Solution Evaluation

Key issue #6: Provisioning of service access based on access token

9.3.3.5 Potential Security Recommendations

- Use OAuth 2.0, as specified in O-RAN Security Protocols Specifications [i.1], clause 4.7 to implement authorization in the R1 interface.
- Enable the SME to act as both an OAuth 2.0 resource and an OAuth 2.0 authorization server.

9.3.4 Solution #4: Secure transmission for client, authorization server and resource server for R1, A1 and O2

9.3.4.1 Background

This solution elaborates on the possibilities to ensure that data for the below flows is secure transmission of request and response between client to authorization server, between client and resource server for R1, A1 and O2.,

1. Between client and authorization server
 - a. The client needs to send the credentials and the authorization request details which need to be secured during transmission.
2. Between client and resource server
 - a. The client needs to send the access token to the authentic resource server which needs to be secured during transmission.

9.3.4.2 Solution Description

Table 9.3.4-1 use case of secure data transmission between client and authorization server

Use Case Stage	Evolution / Specification	<<Uses>> Related use
Goal	1) Transfer the data from client to authorization server securely	
Actors and Roles	Below are examples of HTTP clients - rApp in the role of R1 Service Consumer - SMO towards near RT RIC in the role of A1 consumer - O-RAN assets towards O-cloud in the role of O2 consumer	
Assumptions	TLS is the underlying secure protocol on which the below is built upon.	
Preconditions	The consumer is registered with valid client_id and client_secret The authorization server has a valid X.509 certificate The consumer has the trust anchor for the authorization server's X.509 certificate	
Begins when	The consumer needs to contact the authorization server to send an authorization request	
Step 1 (M)	The client authenticates to the authorization server	
Step 2 (M)	The client sends the request details to authorization server after successful authentication	
Ends when	The consumer receives the access token from the authorization server	
Exceptions	n/a	
Post Conditions	The consumer is in possession of the access token securely	
Traceability	n/a	

```
@startuml
hide footbox
skinparam sequenceActorBackgroundColor White
skinparam sequenceActorBorderColor Black
skinparam sequenceParticipantBackgroundColor White
skinparam sequenceParticipantBorderColor Black
skinparam sequenceArrowColor Black
skinparam sequenceLifeLineBackgroundColor White
skinparam sequenceLifeLineBorderColor Black
skinparam shadowing false
skinparam NoteBackgroundColor White
skinparam NoteBorderColor Black
```

```

Title Secure transmission between client and authorization server
Actor "client" as client
participant "OAuth Authorization Server" as AS

client -> AS:authenticate with client_id and client_secret
client -> AS:send authorization Request
AS -> client: send access token

@enduml

```

Secure transmission between client and authorization server

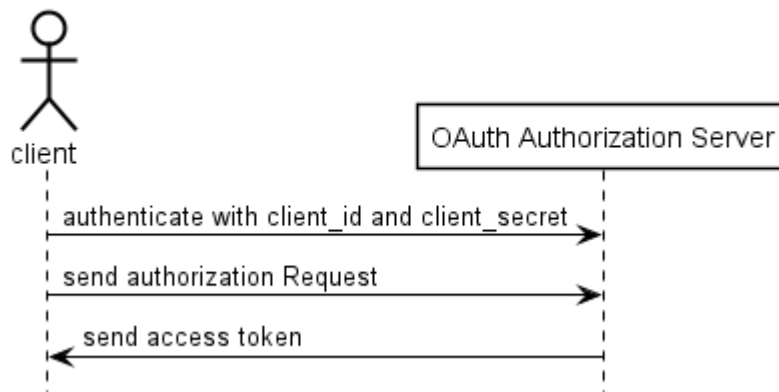


Figure 9.3.4-1 Transmission between client and authorization server

```

@startuml
hide footbox
skinparam sequenceActorBackgroundColor White
skinparam sequenceActorBorderColor Black
skinparam sequenceParticipantBackgroundColor White
skinparam sequenceParticipantBorderColor Black
skinparam sequenceArrowColor Black
skinparam sequenceLifeLineBackgroundColor White
skinparam sequenceLifeLineBorderColor Black
skinparam shadowing false
skinparam NoteBackgroundColor White
skinparam NoteBorderColor Black

Title Secure transmission between client and resource server
Actor "client" as client
participant "resource server" as RS

client <-> RS:mutual authentication
client -> RS:send access token as part of API operation

@enduml

```

Secure transmission between client and resource server

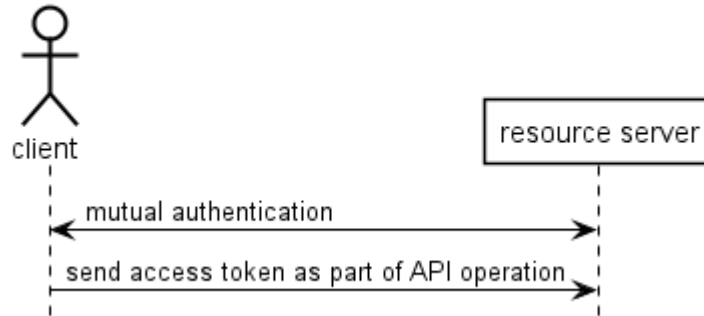


Figure 9.3.4-2 Transmission between client and resource server

9.3.4.2.1 Authentication methods between client, authorization server and resource server

The solutions below are various options to transfer the information securely between the O-RAN OAuth clients and the various servers in the interaction including the authorization server and the various O-RAN resource servers. Below is a summary:

Table 9.3.4-2 Authentication methods

OAuth token end point authentication method	Reference	Short description
Tls_client_auth	RFC 8705 [i.6]	Usage of certificates for both client and server authentication. Usage of certificate hash for certificate bound access tokens
Private_key_JWT (with JWE)	RFC 8705 [i.6]	Usage of certificates for both integrity and confidentiality protection (encryption)
Private_key_JWT	OpenID foundation [i.21]	Usage of certificates for integrity protection
Client_secret_JWT	OpenID foundation [i.21]	Usage of shared secret for example client_secret for integrity protection
Client_secret_basic	RFC 7591 [i.22]	Client_id and client_secret transmission in the header
Client_secret_post	RFC 7591 [i.22]	Client_id and client_secret transmission in the payload
Self_signed_tls_client_auth	RFC 8705 [i.6]	Usage of client side self signed certificates in mutual authentication

Below are the details of the methods explained above

- 1) Tls_client_auth
 - a. Part of RFC 8705.
 - b. Mutual authentication of both client and server. Here certificates are issued by a certificate authority. Also supports token binding.
 - c. For certificate bound access tokens every party proves that they are in possession of their private key.
 - d. Who is in possession of the private key used to establish the connection can use the access token (puts the hash of the certificate (cnf) that has been used).

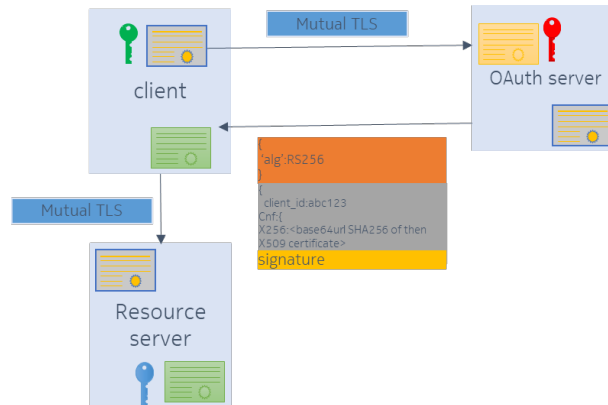


Figure 9.3.4-3 - Tls_client_OAuth method

2) Private_key_JWT (with JWE)

- Part of OpenID foundation specification.
- Public and private key is used to sign and encrypt the JWT token using asymmetric cryptography.
- Contents of the message cannot be seen as they are encrypted.

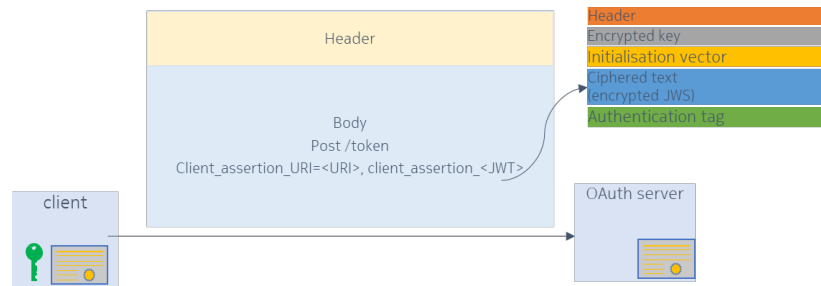


Figure 9.3.4-4 Private_key_JWT with JWE method

3) Private_key_JWT

- Part of OpenID foundation specification
- The client_id and client_secret is part of the JWT token which is integrity protected. The JWT is a string which is attached as a client assertion. Here we AVOID putting the secret in message, instead in the JWT token.
- Confidential client create JSON Web Token along with the public and private key. Hence structure of the JWT is the header, body and the Signature. Hence JWS client has the private key and shares the public key to the authorization server for validation of the request. Hence the contents of the message can be seen but secure from manipulation.

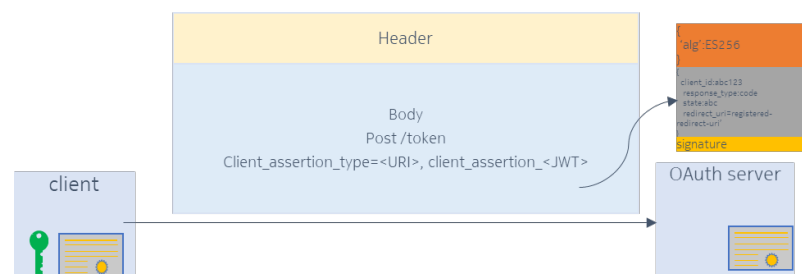


Figure 9.3.4-5 Private_key_JWT method

4) Client_secret_JWT

- a. Part of OpenID foundation specification.
- b. Client_secret_JWT - client_id and client_secret in the JWT token which is integrity protected with a shared secret key (client_secret known to both parties) using HMAC. The JWT is a string which is attached as a client assertion. Here we AVOID putting the secret in message. Instead in the JWT token which is with a MAC (hash function+Secret).

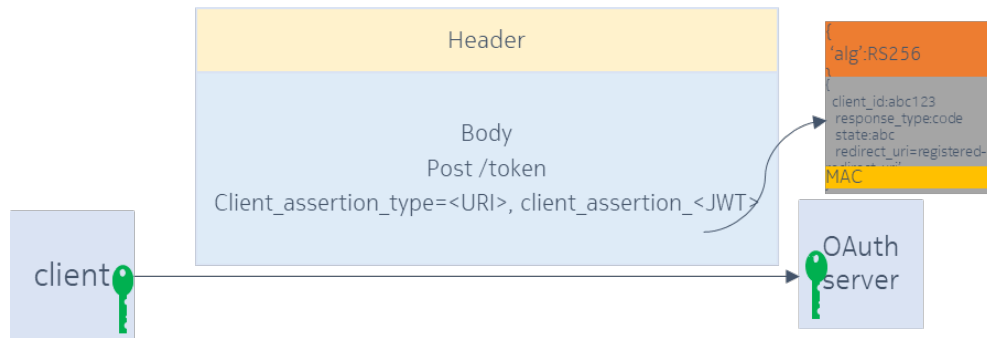


Figure 9.3.4-6 Client_secret_JWT method

5) Client_secret_basic

- a. client_id and client_secret in the header of the message.
- b. Defined in RFC 7591.

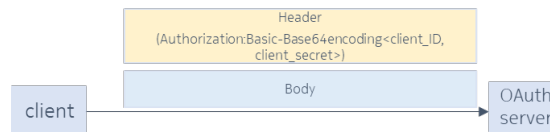


Figure 9.3.4-7 Client_secret_basic method

6) Client_secret_post

- a. client_id and client_secret in the body of the message.
- b. Defined in RFC 7591.

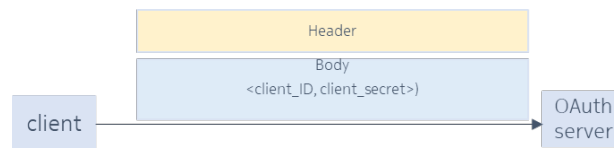


Figure 9.3.4-8 Client_secret_post method

7) Self_signed_tls_client_auth

- a. RFC 8705.
- b. Mutual authentication using self signed certificates. Here certificate is created by self signing.

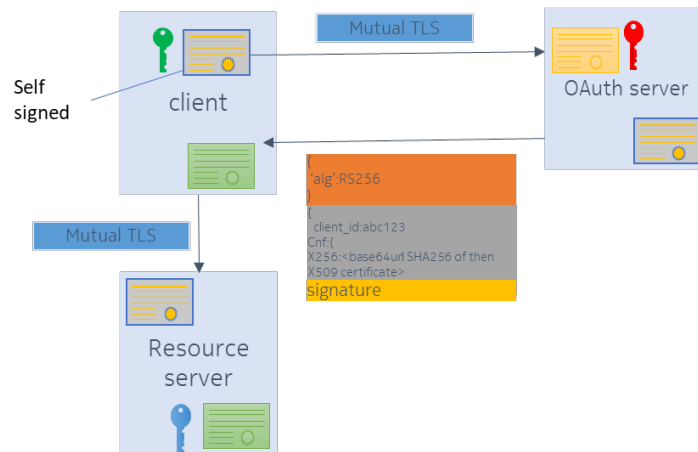


Figure 9.3.4-9 Self_signed_tls_client_auth method

9.3.4.3 Impacts on Existing O-RAN Architecture Elements and Interfaces

R1 client, A1 client and the O2 client needs to choose the authentication method from the options explained above depending on the client and authorization server capabilities:

9.3.4.4 Solution Evaluation

There is a need to support secure transmission of request and response between client and authorization server, between client to resource server. Below is the order of preference of various methods explained above in the order of most preferred to least preferred depending on the authorization server capabilities as well as the security level preferred.

Table 9.3.4-3 Authentication method preferences

OAuth token end point authentication method	Evaluation
Tls_client_auth	Highly preferred
Private_key_JWT (with JWE)	Preferred
Private_key_JWT	Preferred
Client_secret_JWT	Preferred
Client_secret_basic	Preferred
Client_secret_post	Preferred
Self_signed_tls_client_auth	Least preferred

9.3.4.5 Potential Security Recommendations

Editor's note: The security recommendation will be left for future study.

9.3.5 Solution #5: Support for RBAC

9.3.5.1 Background

- Role based access control is an important concept for the authorization mechanism in O-RAN to support either coarse grained or fine-grained access.

- The protected resources expose various operations which can be protected by access rules are mapped to different roles in an operator setup. An operator should assign roles to users (human and non-human) using identity management workflows.

This solution elaborates on the possibilities to

- Explain the association between the users, roles and the access rules (permissions) as defined in [i.19] clause 7.

NOTE: the reference of the 3GPP specification in [i.19] can be considered as an example and customised for the O-RAN interfaces.

9.3.5.2 Solution Description

Table 9.3.5-1 use case of RBAC

Use Case Stage	Evolution / Specification	<<Uses>> Related use
Goal	Design and deploy the role-based access model in an OAuth server for the O-RAN resources	
Actors and Roles	Admin resource server – design of operations on resource server for access control Admin – Network administrator for access control Authorization server	
Assumptions	Resources require coarse and fine grained permissions (access rules) to be designed The solution considers the possibility of adding the permissions manually. There is a possibility that this could also be automated	
Preconditions	n/a	
Begins when	Resource server needs to support access control	
Step 1 (M)	Design coarse grained and fine-grained permissions	
Step 2(M)	Express the permissions of a resource server	
Step 3(M)	Deploy permissions	
Step 4(M)	Deploy roles	
Step 5(M)	Deploy client_ID and its client_secret	
Ends when	Administration of RBAC is completed and resolved in the OAuth server	
Exceptions	n/a	
Post Conditions	n/a	
Traceability	n/a	

```

@startuml
hide footbox
skinparam sequenceActorBackgroundColor Whiteskinparam sequenceActorBorderColor Black
skinparam sequenceParticipantBackgroundColor White
skinparam sequenceParticipantBorderColor Black
skinparam sequenceArrowColor Black
skinparam sequenceLifeLineBackgroundColor White
skinparam sequenceLifeLineBorderColor Black
skinparam shadowing false
skinparam NoteBackgroundColor White
skinparam NoteBorderColor Black

```

```

Actor "Admin Resource server" as AR
Actor "Admin"
participant "Authorization Server" as AS

group Design of permissions \n(resources and operations)
AR-> AR: 1: Design coarse grained and fine grained permissions
AR-> Admin: 2: Express the permissions of a resource server
end

group Provisioning of RBAC model data
Admin -> AS: 3: Deploy permissions
Admin -> AS: 4: Deploy roles
Admin -> AS: 5: Deploy client_ID and its Client_secret
end

@enduml

```

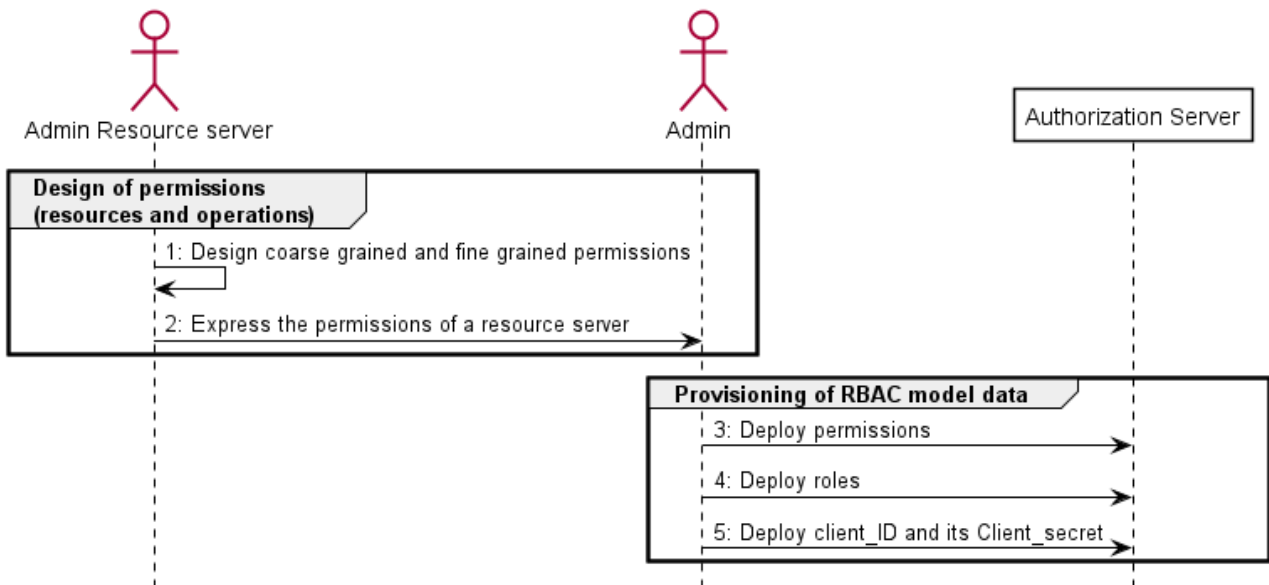


Figure 9.3.5-1 RBAC for client credentials

9.3.5.3 Impacts on Existing O-RAN Architecture Elements and Interfaces

- The O-RAN interfaces should factor in the permissions which should be used in the context of scope in the OAuth flows.
- The keyword scope is used both in an OAuth access token as well as in the OpenAPI to define what a particular interface is allowed to do.

9.3.5.4 Solution Evaluation

- Permissions have to be considered by all the O-RAN APIs,
- For the OpenAPIs this should be expressed as scopes,
- The semantics of the permissions which should be evaluated and validated by the resource server need to be defined.

9.3.5.5 Potential Security Recommendations

Editor's note: The security recommendation will be left for future study.

9.3.6 Solution #6: Profile for client credentials grant type clients

9.3.6.1 Background

- To ensure interoperability we propose the need for prescriptive profiles for a client which need to carry out access control when communicating from one machine to another machine which is the resource server.
 - This solution considers the below aspects for the profile,
 - Resource server which exposes the APIs.
 - Authorization server client (confidential client). Example of confidential client is A1 client interacting with the Near RT RIC platform.
 - Security options with certificate binding [i.6].

9.3.6.2 Solution Description

Table 9.3.6-1 client credentials grant type client profile

Use Case Stage	Evolution / Specification	<<Uses>> Related use
Goal	Derive an OAuth related profile for the machine-to-machine clients	
Actors and Roles	Resource server – provider of the APIs for the client to access Authorization server – an OAuth server Confidential client – a noninteractive client accessing an API	
Assumptions	- The solution considers the possibility of automation - The clients refer to machines and not human users - In registration of client dynamically the initial access to the OAuth server is with the initial access token	
Preconditions	P1. Design of scopes which relates to 4.7.3.2.2. in [i.1] P2. Administration of RBAC (solution related to RBAC) P3. Publish endpoint locations and authorization server capabilities [i.23], clause 2 P4. Client X.509 certificate enrolment	
Begins when	Pre-conditions are met	
Step 1 (M)	Discover the capabilities of authorization server [i.23], clause 2	
Step 2(M)	Register client dynamically [i.22], clause 3.1	
Step 3(M)	Client credentials request (access token lifetime with 'expires_in' parameter) [i.5], clause 4.2.2	
Step 4(M)	Return token [IETF RFC 6750] with certificate binding [i.6], clause 3	
Step 5(M)	API invocation on resource server (access token with certificate binding)	
Ends when	API is accessed with the correct privileges	
Exceptions	n/a	
Post Conditions	n/a	
Traceability	n/a	

NOTE: The design of scopes which is mentioned in the precondition needs further study.

```

@startuml
hide footbox
skinparam sequenceActorBackgroundColor White
skinparam sequenceActorBorderColor Black
skinparam sequenceParticipantBackgroundColor White
skinparam sequenceParticipantBorderColor Black
  
```

```
skinparam sequenceArrowColor Black
skinparam sequenceLifeLineBackgroundColor White
skinparam sequenceLifeLineBorderColor Black
skinparam shadowing false
skinparam NoteBackgroundColor White
skinparam NoteBorderColor Black

participant "Confidential Client" as C
participant "Resource Server" as RS
participant "Authorization Server" as AS

== Pre-condition ==
rnote over RS
P1. Design of scopes
End note

rnote over AS
P2. Administration of RBAC
P3. Publish endpoint locations
    and authorization server capabilities
end note

rnote over C
P4. Client X.509 certificate enrollment
End note

== Runtime ==
group Client Interaction
C -> AS: 1. Discover the capabilities of Authorization server
C -> AS: 2. Register client dynamically

loop (for every request)
C -> AS: 3. Client credentials request(access token lifetime with 'expires_in' parameter)
AS -> C: 4. Return token with certificate binding

group Resource Server Interaction
C -> RS: 5. API call (access token with certificate binding)
End
end
@enduml
```

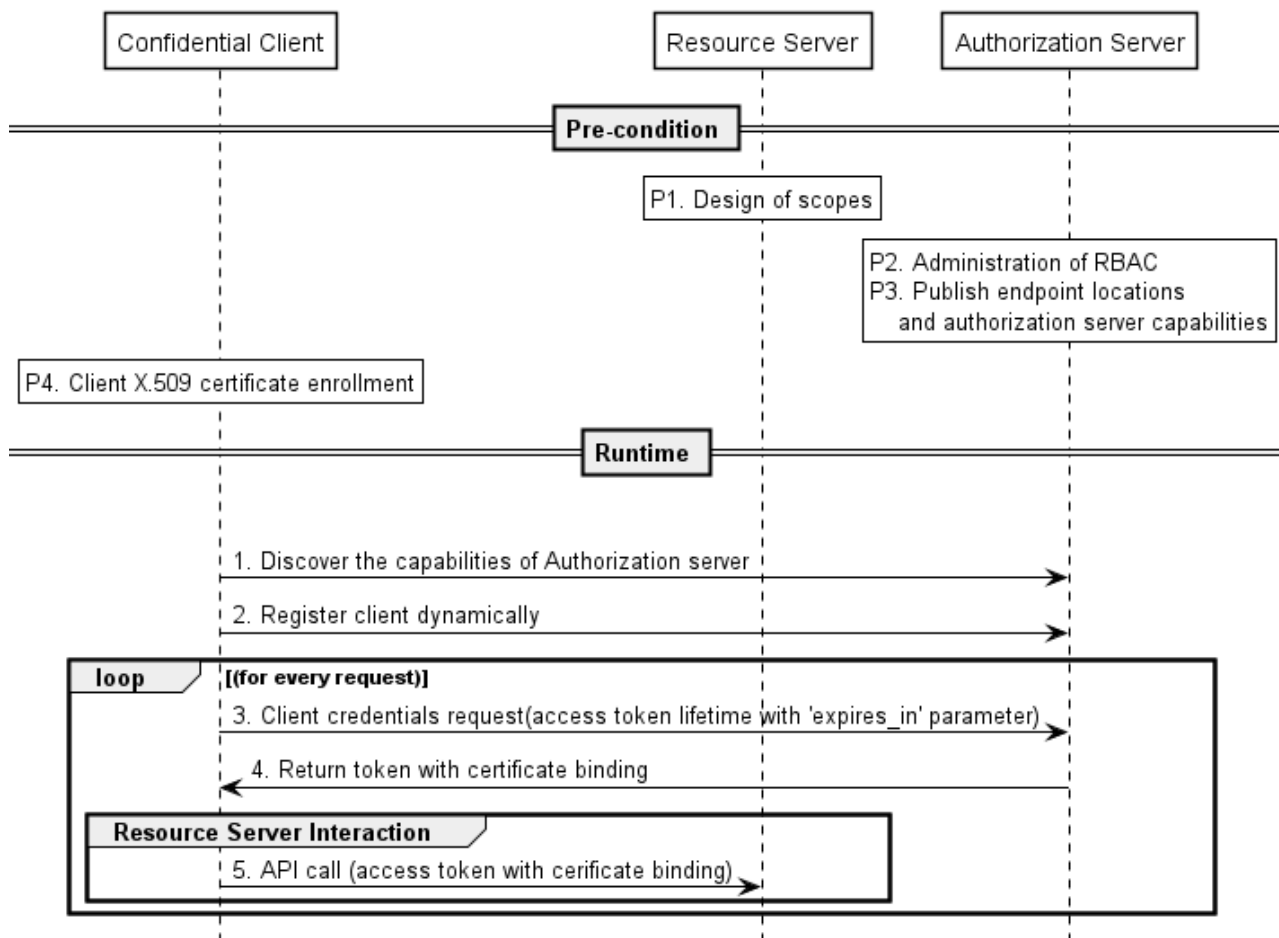


Figure 9.3.6-1 Profile for confidential client

9.3.6.3 Impacts on Existing O-RAN Architecture Elements and Interfaces

Impacts to be considered with respect to the following:

- Resource servers
 - O-RAN related APIs to define the scopes
- Authorization server
 - The OAuth servers should support the following capabilities
 - Token end points support
 - Introspection end point
 - Tokens with reference to access rules
 - Access token lifetime
 - Re-authenticate or refresh tokens
 - Solution options:
 - RBAC deployment
- Client (confidential client)
 - O-RAN related client credentials grant type clients (e.g., xApps) should be capable making a client credential grant request, processing of response and forwarding the access token in an API call.
- Security options
 - O-RAN related http interfaces should support mTLS with certificate binding in [i.6].

9.3.6.4 Solution Evaluation

- To accomplish the access control securely for the O-RAN related machine-to-machine clients a combination of the existing RFCs have to be put together in order to maximise security and also ensure simplicity.

9.3.6.5 Potential Security Recommendations

Editor's note: The security recommendation will be left for future study.

9.3.7 Solution #7: Resolution of authorization decision between authorization server and resource server

9.3.7.1 Background

- In typical operator deployments, RBAC requires that administration and execution of roles (coarse-grained authorization) and permissions (fine-grained authorization) are distributed between the authorization server and resource servers.
- Considering the size of a token and that various APIs have different levels of authorization, the permissions are part of the resource server and not part of the confidential client request.
- The authorization servers are typically centralized overseeing the administration and validation decision of coarse-grained authorization related roles which are part of the confidential client request. The examples of roles could be considered from Appendix I related to the function set group in [i.24].
- The resource servers are typically distributed overseeing the roles to permission mapping administration and validation decision of fine-grained authorization.
- Permissions are considered to be the operation (usually CRUD) on a particular object in the resource server as in [i.19] clause 7.
- In the solution below we consider examples of roles as rol1, rol2, rol3, etc. to be mapped to permissions per1, per2, per3, etc. for ease of understanding.

NOTE 1: The above discussion related to coarse-grained authorization and fine-grained authorization which is mentioned in the 9.3.7 needs further study.

NOTE 2: The above discussion related to roles in [i.24] which is mentioned in the 9.3.7.1 and 9.3.7.3 needs further study.

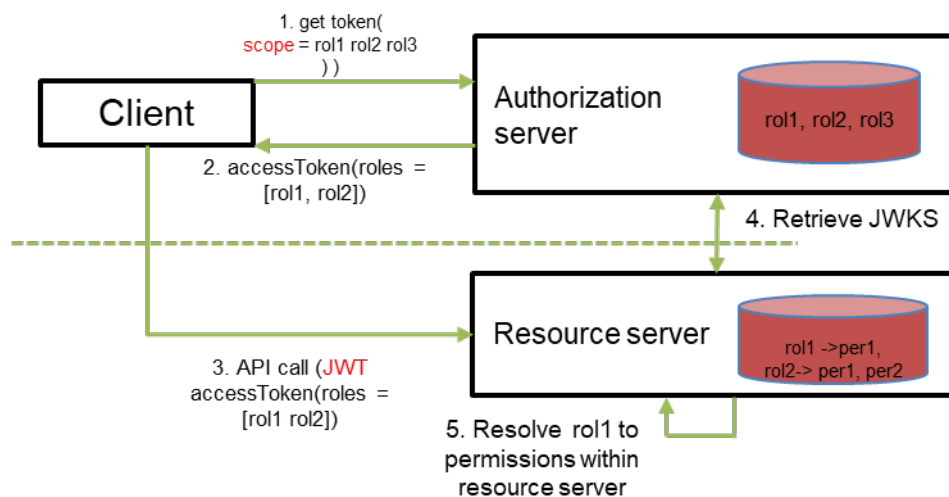


Figure 9.3.7-1 - Validation decision of roles and permissions

This solution elaborates on the possibilities to

- Split the administration of the roles and permissions between the authorization server and resource server.
- Validate the decision of the roles and permissions between the authorization server and resource server.

9.3.7.2 Solution Description

Table 9.3.7-1 use case of authorization decision between authorization server and resource server

Use Case Stage	Evolution / Specification	<<Uses>> Related use
Goal	Resolution of authorization related validation decision to achieve the principle of least privilege	
Actors and Roles	Confidential client – making an API call on a resource server with the assigned role Authorization server – Validating the scope and the roles of client Resource server – Validating the fine-grained authorization after mapping them to the roles received in access token	
Assumptions	Coarse-grained authorization roles are administered in the Authorization server (e.g.: rol1, rol2, rol3, etc) Fine-grained authorization is administered in the resource server (e.g.: per1, per2, per3, etc.). permission defines the granular object in the resource server and operation (CRUD) that can be executed on it Resource server to verify the token for every confidential client during the service initiation	
Preconditions	n/a	
Begins when	Confidential client makes a token request	
Step 1 (M)	Send a token request with requested roles (e.g.: rol1, rol2, rol3) included in the 'scope' attribute	
Step 2(M)	Validate the roles related to the ClientID	
Step 3(M)	Return JWT access token with authorized roles (e.g.: rol1) included in the 'roles' attribute	
Step 4(M)	API call (JWT access token with role (e.g.: rol1))	
Step 5(M)	Check mapping of the role to permission (e.g.: rol1 -> per1)	
Step 6(M)	Validate the permission of the fine-grained authorization operation in the API call	
Ends when	Roles and permissions are resolved by respective entity and API call is executed with required authorization resolution	
Exceptions	n/a	
Post Conditions	n/a	
Traceability	n/a	

```

@startuml
hide footbox
skinparam sequenceActorBackgroundColor White
skinparam sequenceActorBorderColor Black
skinparam sequenceParticipantBackgroundColor White
skinparam sequenceParticipantBorderColor Black
skinparam sequenceArrowColor Black
skinparam sequenceLifeLineBackgroundColor White
skinparam sequenceLifeLineBorderColor Black
skinparam shadowing false
skinparam NoteBackgroundColor White
skinparam NoteBorderColor Black

Title Authorization resolution for fine-grained authorization
participant "Confidential Client" as C
participant "Resource Server" as RS
participant "Authorization Server" as AS

```

```

== Runtime ==
group Client Interaction
C -> AS: 1. Send a token request with requested roles (e.g.:rol1, rol2, rol3..) included in
the 'scope' attribute
AS -> AS:2. Validate the roles related to the clientID
AS -> C: 3. Return JWT access token with authorized roles(e.g.: rol1) included in the 'roles'
attribute

loop (for every request)

C -> RS: 4. API call (JWT access token with role(e.g.: rol1))

RS -> RS:5. Check mapping of the role to permission(e.g.: rol1 -> per1)
RS -> RS:6. Validate the permission of the fine-grained authorization operation in the API
call

End
end
@enduml

```

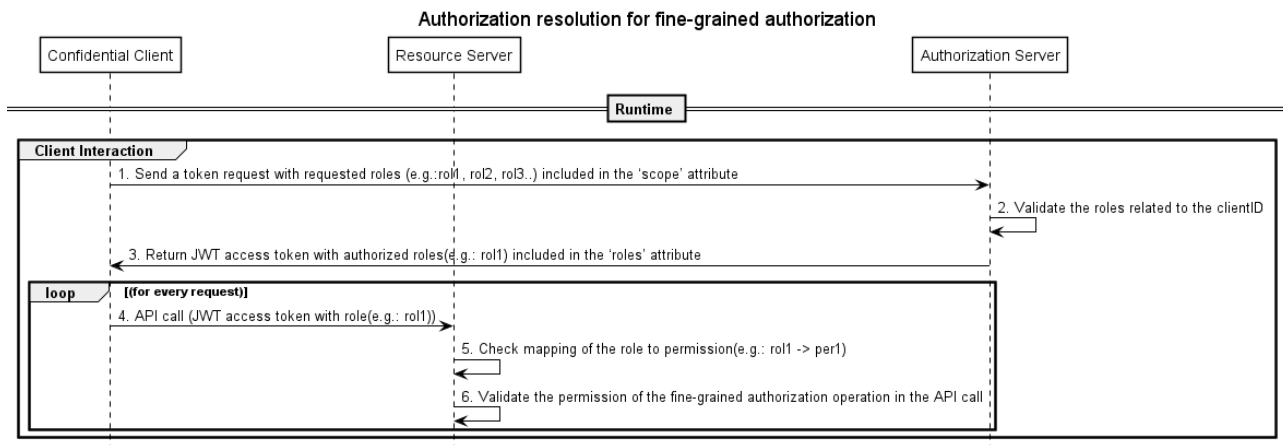


Figure 9.3.7-2 Roles and permission resolution between the authorization server and resource server

9.3.7.3 Impacts on Existing O-RAN Architecture Elements and Interfaces

- O-RAN related confidential clients request access token with the roles as part of the claim. The authorization server administers and validates the roles to client token requests. Examples of roles are defined in function set groups in Appendix I for the network management layer in [i.24].
- The O-RAN interfaces (APIs) may factor in the validation of permissions which are mapped to the roles the access token.

9.3.7.4 Solution Evaluation

- Roles have to be resolved and validated by the authorization server.
- Roles to permission mapping may be administered and validated in the resource server for the various API calls.

9.3.7.5 Potential Security Recommendations

Editor's note: The security recommendation will be left for future study.

10 Study and recommendation an O-RAN Authorization Server architecture

Editor's note: This section will do a study and recommendation on an authorization server architecture to overlay on the O-RAN architecture.

11 Conclusion

Editor's note : This clause will provide the summary for the TR and mapping tables the key issues and solutions and it will submit the recommendations to WG1 for O-RAN authorization server.

Annex:Change history/Change request (history)

Date	Revision	Description
2023.05.10	00.00.01	Initial TS skeleton
2023.08.23	00.00.02	Updated the scope and reference for the TR
2023.10.16	00.00.03	Updated the TR based on CR approved in OAuth 2.0 meeting for Nov train
2023.10.24	00.00.04	Updated the TR based on CR approved during O-RAN F2F phoenix meeting
2023.11.21	01.00	Final version 01.00
2024.02.07	01.00.01	Updated the TR based on CR approved for OAuth 2.0 TR WI meeting in Dec 2023 and Jan 2024
2024.03.05	01.00.02	Updated the TR based on CR approved for OAuth 2.0 TR WI meeting in Feb 2024
2024.06.26	02.00.01	Updated the TR based on CR approved for OAuth 2.0 TR WI meeting from April to June 2024
2024.07.05	02.00.01	Updated the TR based on the review comments during July 2024 plenary approvals
2024.10.30	03.00.01	Updated the TR based CR approved before the WG11 F2F oct 2024 meeting
2024.11.12	03.00.02	Updated the TR based CR approved after the WG11 F2F oct 2024 meeting
2024.11.26	03.00.03	Updated the TR based on editorial comments during the WG11 plenary approval for Nov 2024 train
2024.11.27	03.00.04	Updated the TR based on editorial comments (OAuthTR meeting update) during the WG11 plenary approval for Nov 2024 train
2025.03.04	04.01	Updated the TR based on CR approved for OAuth 2.0 TR WI meeting in Dec 2024 Jan 2025 and Feb 2025
2025.03.04	04.02	Updated the TR based on CR approved in OAuth 2.0 WI meeting on 04/03/2025
2025.03.18	04.03	Updated the TR based on review comments received on or before OAuth 2.0 WI meeting on 18/03/2025
2025.03.25	05.00.01	Updated the TR based on review comments received during voting of OAuth 2.0 march 2025 train
2025.05.27	05.01	Updated the TR based on below CR approved for OAuth 2.0 TR WI meeting in April 2025 and May 2025 ERI-2025.03.25-WG11-CR-224-OAuth2.0-Clause2Considerationupdates-v02.docx ERI-2025.01.04-WG11-CR0225-OAuth2.0-Clause7-Risk Assessment_Update(1)_v04.docx ERI-2025.04.15-WG11-CR0233-OAuth2.0-Clause5.5SecurityControlsupdates-v03.docx ERI-2025.04.15-WG11-CR0234-OAuth2.0-Clause5.1AssetsIDupdates-v01.docx ERI-2025.04.29-WG11-CR0236-OAuth2.0-Clause7-Risk Assessment_Update(2)_v02.docx ERI-2025.05.13-WG11-CR0240-OAuth2.0-Clause7.2-Risk Assessment_Update(3)_v01.docx ERI-2025.05.27-WG11-CR0245-OAuth2.0-Clause7.2-Risk Assessment_Update(4)_v01.docx
2025.06.10	05.02	Updated the TR based on review comments received during the OAuth2.0 TR meeting 60 on 10/06/2025